

Chapter 1

Introduction

1.1 Introduction

Optimization Theory and Methods is a young subject in applied mathematics, computational mathematics and operations research which has wide applications in science, engineering, business management, military and space technology. The subject is involved in optimal solution of problems which are defined mathematically, i.e., given a practical problem, the “best” solution to the problem can be found from lots of schemes by means of scientific methods and tools. It involves the study of optimality conditions of the problems, the construction of model problems, the determination of algorithmic method of solution, the establishment of convergence theory of the algorithms, and numerical experiments with typical problems and real life problems. Though optimization might date back to the very old extreme-value problems, it did not become an independent subject until the late 1940s, when G.B. Dantzig presented the well-known simplex algorithm for linear programming. After the 1950s, when conjugate gradient methods and quasi-Newton methods were presented, the nonlinear programming developed greatly. Now various modern optimization methods can solve difficult and large scale optimization problems, and become an indispensable tool for solving problems in diverse fields.

The general form of optimization problems is

$$\begin{array}{ll} \min & f(x) \\ \text{s.t.} & x \in X, \end{array} \quad (1.1.1)$$

where $x \in R^n$ is a decision variable, $f(x)$ an objective function, $X \subset R^n$

a constraint set or feasible region. Particularly, if the constraint set $X = R^n$, the optimization problem (1.1.1) is called an unconstrained optimization problem:

$$\min_{x \in R^n} f(x). \quad (1.1.2)$$

The constrained optimization problem can be written as follows:

$$\begin{aligned} \min_{x \in R^n} \quad & f(x) \\ \text{s.t.} \quad & c_i(x) = 0, \quad i \in E, \\ & c_i(x) \geq 0, \quad i \in I, \end{aligned} \quad (1.1.3)$$

where E and I are, respectively, the index set of equality constraints and inequality constraints, $c_i(x)$, ($i = 1, \dots, m \in E \cup I$) are constraint functions. When both objective function and constraint functions are linear functions, the problem is called linear programming. Otherwise, the problem is called nonlinear programming.

This book mainly studies solving unconstrained optimization problem (1.1.2) and constrained optimization problem (1.1.3) from the view points of both theory and numerical methods. Chapters 2 to 7 deal with unconstrained optimization. Chapters 8 to 13 discuss constrained optimization. Finally, in Chapter 14, we give a simple and comprehensive introduction to nonsmooth optimization.

1.2 Mathematics Foundations

In this section, we shall review a number of results from linear algebra and analysis which are useful in optimization theory and methods.

Throughout this book, R^n will denote the real n -dimensional linear space of column vector x with components $x_1, \dots, x_n$, and C^n the corresponding space of complex column vectors. For $x \in R^n$, x^T denotes the transpose of x , while, for $x \in C^n$, x^H is the conjugate transpose. A real $m \times n$ matrix $A = (a_{ij})$ defines a linear mapping from R^n to R^m and will be written as $A \in R^{m \times n}$ or $A \in L(R^n, R^m)$ to denote either the matrix or the linear operator. Similarly, a complex $m \times n$ matrix A will be written as $A \in C^{m \times n}$ or $A \in L(C^n, C^m)$.

1.2.1 Norm

Definition 1.2.1 A mapping $\|\cdot\|$ is called a norm if and only if it satisfies the following properties:

- (i) $\|x\| \geq 0, \forall x \in R^n; \|x\| = 0$ if and only if $x = 0$;
- (ii) $\|\alpha x\| = |\alpha|\|x\|, \forall \alpha \in R, x \in R^n$;
- (iii) $\|x + y\| \leq \|x\| + \|y\|, \forall x, y \in R^n$.

Well-known examples of vector norm are as follows:

$$\|x\|_\infty = \max_{1 \leq i \leq n} |x_i|, \quad (l_\infty\text{-norm}) \quad (1.2.1)$$

$$\|x\|_1 = \sum_{i=1}^n |x_i|, \quad (l_1\text{-norm}) \quad (1.2.2)$$

$$\|x\|_2 = \left(\sum_{i=1}^n |x_i|^2 \right)^{1/2}, \quad (l_2\text{-norm}). \quad (1.2.3)$$

The above examples are particular cases of l_p -norm which is defined as

$$\|x\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}, \quad (l_p\text{-norm}). \quad (1.2.4)$$

Another vector norm frequently used is the ellipsoid norm which is defined as

$$\|x\|_A = (x^T A x)^{1/2}, \quad (1.2.5)$$

where $A \in R^{n \times n}$ is a symmetric and positive definite matrix.

Similarly, we can define a matrix norm.

Definition 1.2.2 Let $A, B \in R^{m \times n}$. A mapping $\|\cdot\| : R^{m \times n} \rightarrow R$ is said to be a matrix norm if it satisfies the properties

- (i) $\|A\| \geq 0, \forall A \in R^{m \times n}; \|A\| = 0$ if and only if $A = 0$;
- (ii) $\|\alpha A\| = |\alpha|\|A\|, \forall \alpha \in R, A \in R^{m \times n}$;
- (iii) $\|A + B\| \leq \|A\| + \|B\|, \forall A, B \in R^{m \times n}$.

Corresponding to the above vector l_p -norm, we have the matrix l_p -norm:

$$\|A\|_p = \sup_{x \neq 0} \frac{\|Ax\|_p}{\|x\|_p} = \max_{\|x\|_p=1} \|Ax\|_p \quad (1.2.6)$$

which is said to be induced by, or subordinate to, the vector l_p -norm. In particular,

$$\|A\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^m |a_{ij}|, \quad (\text{maximum column norm}) \quad (1.2.7)$$

$$\|A\|_\infty = \max_{1 \leq i \leq m} \sum_{j=1}^n |a_{ij}|, \quad (\text{maximum row norm}) \quad (1.2.8)$$

$$\|A\|_2 = \left(\lambda_{\max}(A^T A) \right)^{1/2}, \quad (\text{spectral norm.}) \quad (1.2.9)$$

Obviously, we have

$$\|A^{-1}\|_p = \frac{1}{\min_{x \neq 0} \frac{\|Ax\|_p}{\|x\|_p}}.$$

For an induced matrix norm, we always have $\|I\| = 1$, where I is an $n \times n$ identity matrix. More generally, for any vector norm $\|\cdot\|_\alpha$ on R^n and $\|\cdot\|_\beta$ on R^m , the matrix norm is defined by

$$\|A\|_{\alpha,\beta} = \sup_{x \neq 0} \frac{\|Ax\|_\beta}{\|x\|_\alpha}. \quad (1.2.10)$$

The most frequently used matrix norms also include the Frobenius norm

$$\|A\|_F = \left(\sum_{i=1}^m \sum_{j=1}^n |a_{ij}|^2 \right)^{1/2} = [\text{tr}(A^T A)]^{1/2}, \quad (1.2.11)$$

where $\text{tr}(\cdot)$ denotes the trace of a square matrix with $\text{tr}(A) = \sum_{i=1}^n a_{ii}$. The trace satisfies

1. $\text{tr}(\alpha A + \beta B) = \alpha \text{tr}(A) + \beta \text{tr}(B)$;
2. $\text{tr}(A^T) = \text{tr}(A)$;
3. $\text{tr}(AB) = \text{tr}(BA)$;
4. $\text{tr}(A) = \sum_{i=1}^n \lambda_i$ if the eigenvalues of A are denoted by $\lambda_1, \dots, \lambda_n$.

The weighted Frobenius norm and weighted l_2 -norm are defined, respectively, as

$$\|A\|_{M,F} = \|MAM\|_F, \quad \|A\|_{M,2} = \|MAM\|_2, \quad (1.2.12)$$

where M is an $n \times n$ symmetric and positive definite matrix.

Further, let $A \in R^{n \times n}$; if we define $\|x\|' = \|Px\|$ for all $x \in R^n$ and P an arbitrary nonsingular matrix, then

$$\|A\|' = \|PAP^{-1}\|. \quad (1.2.13)$$

The orthogonally invariant matrix norm is a class of important norms which satisfies, for $A \in R^{m \times n}$ and U an $m \times m$ orthogonal matrix, the identity

$$\|UA\| = \|A\|. \quad (1.2.14)$$

Clearly, the l_2 -norm and the Frobenius norm are orthogonally invariant matrix norms.

A vector norm $\|\cdot\|$ and a matrix norm $\|\cdot\|'$ are said to be consistent if, for every $A \in R^{m \times n}$ and $x \in R^n$,

$$\|Ax\| \leq \|A\|' \|x\|. \quad (1.2.15)$$

Obviously, the l_p -norm has this property, i.e.,

$$\|Ax\|_p \leq \|A\|_p \|x\|_p. \quad (1.2.16)$$

More generally, for any vector norm $\|\cdot\|_\alpha$ on R^n and $\|\cdot\|_\beta$ on R^m we have

$$\|Ax\|_\beta \leq \|A\|_{\alpha,\beta} \|x\|_\alpha, \quad (1.2.17)$$

where $\|A\|_{\alpha,\beta}$ is defined by

$$\|A\|_{\alpha,\beta} = \sup_{x \neq 0} \frac{\|Ax\|_\beta}{\|x\|_\alpha} \quad (1.2.18)$$

which is subordinate to the vector norm $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$.

Likewise, if a norm $\|\cdot\|$ satisfies

$$\|AB\| \leq \|A\| \|B\|, \quad (1.2.19)$$

we say that the matrix norm satisfies the consistency condition (or submultiplicative property). It is easy to see that the Frobenius norm and the induced matrix norms satisfy the consistency condition, and we have

$$\|AB\|_F \leq \min\{\|A\|_2 \|B\|_F, \|A\|_F \|B\|_2\}. \quad (1.2.20)$$

Next, about the equivalence of norms, we have

Definition 1.2.3 *Let $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$ be two arbitrary norms on R^n . If there exist $\mu_1, \mu_2 > 0$, such that*

$$\mu_1 \|x\|_\alpha \leq \|x\|_\beta \leq \mu_2 \|x\|_\alpha, \quad \forall x \in R^n, \quad (1.2.21)$$

we say that the norms $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$ are equivalent.

In particular, we have

$$\|x\|_2 \leq \|x\|_1 \leq \sqrt{n} \|x\|_2, \quad (1.2.22)$$

$$\|x\|_\infty \leq \|x\|_2 \leq \sqrt{n} \|x\|_\infty, \quad (1.2.23)$$

$$\|x\|_\infty \leq \|x\|_1 \leq n \|x\|_\infty, \quad (1.2.24)$$

$$\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1, \quad (1.2.25)$$

$$\sqrt{\lambda} \|x\|_2 \leq \|x\|_A \leq \sqrt{\Lambda} \|x\|_2, \quad (1.2.26)$$

where λ and Λ are the smallest and the largest eigenvalues of A respectively. For $A \in R^{m \times n}$, we have

$$\|A\|_2 \leq \|A\|_F \leq \sqrt{n} \|A\|_2, \quad (1.2.27)$$

$$\max_{i,j} |a_{ij}| \leq \|A\|_2 \leq \sqrt{mn} \max_{i,j} |a_{ij}|, \quad (1.2.28)$$

$$\frac{1}{\sqrt{n}} \|A\|_\infty \leq \|A\|_2 \leq \sqrt{m} \|A\|_\infty, \quad (1.2.29)$$

$$\frac{1}{\sqrt{m}} \|A\|_1 \leq \|A\|_2 \leq \sqrt{n} \|A\|_1. \quad (1.2.30)$$

By use of norms, it is immediate to introduce the notation of distance. Let $x, y \in R^n$, the distance between two points x and y is defined by $\|x - y\|$. In particular, in the 2-norm, if $x = (x_1, \dots, x_n)^T, y = (y_1, \dots, y_n)^T$, then

$$\|x - y\|_2 = \sqrt{(x_1 - y_1)^2 + \dots + (x_n - y_n)^2}$$

which is just a direct generalization of distance in analytical geometry.

Obviously, by Definition 1.2.1, we have the following properties of distance:

1. $\|x - y\| \geq 0, \|x - y\| = 0$ if and only if $x = y$.

$$2. \|x - z\| \leq \|x - y\| + \|y - z\|.$$

$$3. \|x - y\| = \|y - x\|.$$

A vector sequence $\{x_k\}$ is said to be convergent to x^* if

$$\lim_{k \rightarrow \infty} \|x_k - x^*\| = 0. \quad (1.2.31)$$

A matrix sequence $\{A_k\}$ is said to be convergent to A if

$$\lim_{k \rightarrow \infty} \|A_k - A\| = 0. \quad (1.2.32)$$

Choice of norms is irrelevant since all norms in finite dimension space are equivalent.

Definition 1.2.4 A sequence $\{x_k\} \subset R^n$ is said to be a Cauchy sequence if

$$\lim_{m, l \rightarrow \infty} \|x_m - x_l\| = 0; \quad (1.2.33)$$

i.e., given $\epsilon > 0$, there is an integer N such that $\|x_m - x_l\| < \epsilon$ for all $m, l > N$.

In R^n , a sequence $\{x_k\}$ converges if and only if the sequence $\{x_k\}$ is a Cauchy sequence. However, in a normed space, a Cauchy sequence may not be convergent.

We conclude this subsection with several inequalities on norms.

(1) Cauchy-Schwarz inequality :

$$|x^T y| \leq \|x\|_2 \|y\|_2, \quad (1.2.34)$$

the equality holds if and only if x and y are linearly dependent.

(2) Let A be an $n \times n$ symmetric and positive definite matrix, then the inequality

$$|x^T A y| \leq \|x\|_A \|y\|_A \quad (1.2.35)$$

holds; the equality holds if and only if x and y are linearly dependent.

(3) Let A be an $n \times n$ symmetric and positive definite matrix, then the inequality

$$|x^T y| \leq \|x\|_A \|y\|_{A^{-1}} \quad (1.2.36)$$

holds; the equality holds if and only if x and $A^{-1}y$ are linearly dependent.

(4) Young inequality: Assume that real numbers p and q are each larger than 1, and $\frac{1}{p} + \frac{1}{q} = 1$. If x and y are also real numbers, then

$$xy \leq \frac{x^p}{p} + \frac{y^q}{q}, \quad (1.2.37)$$

and equality holds if and only if $x^p = y^q$.

Proof. Set $s = x^p$ and $t = y^q$. From the arithmetic-geometry inequality, we immediately have

$$xy = s^{1/p} t^{1/q} \leq \frac{s}{p} + \frac{t}{q} = \frac{x^p}{p} + \frac{y^q}{q}.$$

Further, the equality holds if and only if $s = t$, i.e., $x^p = y^q$. $\square$

(5) Hölder inequality:

$$|x^T y| \leq \|x\|_p \|y\|_q = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p} \left(\sum_{i=1}^n |y_i|^q \right)^{1/q}, \quad (1.2.38)$$

where p and q are real numbers larger than 1 and satisfy $\frac{1}{p} + \frac{1}{q} = 1$.

Proof. If $x = 0$ or $y = 0$, the result is trivial. Now we assume that both x and y are not zero. From Young inequality, we have

$$\frac{|x_i y_i|}{\|x\|_p \|y\|_q} \leq \frac{1}{p} \frac{|x_i|^p}{\|x\|_p^p} + \frac{1}{q} \frac{|y_i|^q}{\|y\|_q^q}, \quad i = 1, \dots, n.$$

Taking the sum over i on both sides of the above inequality yields

$$\begin{aligned} & \frac{1}{\|x\|_p \|y\|_q} \sum_{i=1}^n |x_i y_i| \\ & \leq \frac{1}{p \|x\|_p^p} \sum_{i=1}^n |x_i|^p + \frac{1}{q \|y\|_q^q} \sum_{i=1}^n |y_i|^q \\ & = \frac{1}{p} + \frac{1}{q} \\ & = 1. \quad \square \end{aligned}$$

Multiplying $\|x\|_p \|y\|_q$ on both sides gives our result.

(6) Minkowski inequality:

$$\|x + y\|_p \leq \|x\|_p + \|y\|_p, \quad (1.2.39)$$

i.e.,

$$\left(\sum_{i=1}^n |x_i + y_i|^p \right)^{1/p} \leq \left(\sum_{i=1}^n |x_i|^p \right)^{1/p} + \left(\sum_{i=1}^n |y_i|^p \right)^{1/p}, \quad (1.2.40)$$

where $p \geq 1$. The proof of this inequality will be given in §1.3.2 as an application of the convexity of a function.

1.2.2 Inverse and Generalized Inverse of a Matrix

In this subsection we collect some basic results of inverse and generalized inverse.

Theorem 1.2.5 (*Von-Neumann Lemma*) *Let $\|\cdot\|$ be a consistent matrix norm with $\|I\| = 1$. Let $E \in R^{n \times n}$. If $\|E\| < 1$, then $I - E$ is nonsingular, and*

$$(I - E)^{-1} = \sum_{k=0}^{\infty} E^k, \quad (1.2.41)$$

$$\|(I - E)^{-1}\| \leq \frac{1}{1 - \|E\|}. \quad (1.2.42)$$

If $A \in R^{n \times n}$ is nonsingular and $\|A^{-1}(B - A)\| < 1$, then B is nonsingular and satisfies

$$B^{-1} = \sum_{k=0}^{\infty} (I - A^{-1}B)^k A^{-1}, \quad (1.2.43)$$

and

$$\|B^{-1}\| \leq \frac{\|A^{-1}\|}{1 - \|A^{-1}(B - A)\|}. \quad (1.2.44)$$

Proof. Since $\|E\| < 1$, then

$$S_k \triangleq I + E + E^2 + \cdots + E^k$$

defines a Cauchy sequence, and hence S_k is convergent. So,

$$\sum_{k=0}^{\infty} E^k = \lim_{k \rightarrow \infty} S_k = (I - E)^{-1}$$

which proves (1.2.41)-(1.2.42).

Since A is nonsingular and $\|A^{-1}(B - A)\| = \|(I - A^{-1}B)\| < 1$, by setting $E = I - A^{-1}B$ and using (1.2.41) and (1.2.42), we obtain immediately (1.2.43) and (1.2.44). $\square$

This theorem indicates that the matrix B is invertible if B is sufficiently approximate to an invertible matrix A . The above theorem also can be written in the following form which sometimes is said to be the perturbation theorem:

Theorem 1.2.6 *Let $A, B \in R^{n \times n}$. Assume that A is invertible with $\|A^{-1}\| \leq \alpha$. If $\|A - B\| \leq \beta$ and $\alpha\beta < 1$, then B is also invertible, and*

$$\|B^{-1}\| \leq \frac{\alpha}{1 - \alpha\beta}. \quad (1.2.45)$$

Let L and M be subspaces of R^n . The sum of two subspaces L and M is defined as

$$L + M = \{x = y + z \mid y \in L, z \in M\}. \quad (1.2.46)$$

The intersection of two subspaces L and M is defined as

$$L \cap M = \{x \mid x \in L \text{ and } x \in M\}. \quad (1.2.47)$$

Two subspaces L and M are orthogonal, denoted by $L \perp M$, if

$$\langle y, z \rangle = 0, \quad \forall y \in L, \forall z \in M.$$

R^n is said to be a direct sum of L and M , denoted by

$$R^n = L \oplus M,$$

if and only if $R^n = L + M$ and $L \cap M = \{0\}$.

Let $R^n = L \oplus M$. If a linear operator $P : R^n \rightarrow R^n$ satisfies

$$Py = y, \forall y \in L; \quad Pz = 0, \forall z \in M,$$

then P is called a projector of R^n onto the subspace L along the subspace M . Such a projector is denoted by $P_{L,M}$ or P . If $M \perp L$, then the above projector is called an orthogonal projector, denoted by P_L or P .

Normally, $C^{m \times n}$ denotes a set of all complex $m \times n$ matrices, $C_r^{m \times n}$ denotes a set of all complex $m \times n$ matrices with rank r . A^* denotes the conjugate transpose of a matrix A . For a real matrix, $R^{m \times n}$ and $R_r^{m \times n}$ have

similar meaning. Now we present some definitions and representations of the generalized inverse of a matrix A .

Let $A \in C^{m \times n}$. Then $A^+ \in C^{n \times m}$ is a Moore-Penrose generalized inverse of A if

$$AA^+A = A, A^+AA^+ = A^+, (AA^+)^* = AA^+, (A^+A)^* = A^+A, \quad (1.2.48)$$

or equivalently,

$$AA^+ = P_{R(A)}, A^+A = P_{R(A^+)}, \quad (1.2.49)$$

where $P_{R(A)}$ and $P_{R(A^+)}$ are the orthogonal projectors on range $R(A)$ and $R(A^+)$ respectively.

If $A \in C_r^{m \times n}$ and A has the orthogonal decomposition

$$A = Q^*RP, \quad (1.2.50)$$

where Q and P are $m \times m$ and $n \times n$ unitary matrices respectively, $R \in C^{m \times n}$,

$$R = \begin{bmatrix} R_{11} & 0 \\ 0 & 0 \end{bmatrix},$$

where R_{11} is the $r \times r$ nonsingular upper triangular matrix, then

$$A^+ = P^*R^+Q, \quad (1.2.51)$$

where

$$R^+ = \begin{bmatrix} R_{11}^{-1} & 0 \\ 0 & 0 \end{bmatrix}.$$

Similarly, if $A \in C_r^{m \times n}$ has the singular value decomposition (SVD)

$$A = UDV^*, \quad (1.2.52)$$

where U and V are $m \times m$ and $n \times n$ unitary matrices respectively,

$$D = \begin{bmatrix} \Sigma & 0 \\ 0 & 0 \end{bmatrix} \in C^{m \times n},$$

where $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_r)$, $\sigma_i > 0$ ($i = 1, \dots, r$) are the nonzero singular values of A , then

$$A^+ = VD^+U^*, \quad (1.2.53)$$

where

$$D^+ = \begin{bmatrix} \Sigma^{-1} & 0 \\ 0 & 0 \end{bmatrix}.$$

An important role of the generalized inverse is that it offers the solution of general linear equations (including singular, rectangular, or inconsistent case). In the following we state this theorem and prove it by the singular value decomposition.

Theorem 1.2.7 *Let $A \in C^{m \times n}$, $b \in C^m$. Then $\bar{x} = A^+b$ is the unique solution of $Ax = b$, i.e.,*

$$\|\bar{x}\| \leq \|x\|, \forall x \in \{x \mid \|Ax - b\| \leq \|Az - b\|, \forall z \in C^m\}. \quad (1.2.54)$$

Such an $\bar{x}$ is called the minimal least-squares solution of $Ax = b$.

Proof. From the singular value decomposition (1.2.52), (1.2.54) is equivalent to

$$\min_{x \in R^n} \{\|V^*x\| \mid \|DV^*x - U^*b\| \leq \|DV^*z - U^*b\|, \forall z \in R^n\}$$

i.e., for $y = V^*x$,

$$\min_{y \in R^n} \{\|y\| \mid \|Dy - U^*b\| \leq \|D\hat{z} - U^*b\|, \forall \hat{z} \in R^n\}. \quad (1.2.55)$$

Since

$$\|Dy - U^*b\|^2 = \sum_{i=1}^r (\sigma_i y_i - (U^*b)_i)^2 + \sum_{i=r+1}^m ((U^*b)_i)^2$$

which is minimized by any y with $y_i = (U^*b)_i / \sigma_i$, ($i = 1, \dots, r$) and $\|y\|$ is minimized by setting $y_i = 0$ ($i = r+1, \dots, m$), then $y = D^+U^*b$ is the minimal least-squares solution of (1.2.55). Therefore $\bar{x} = VD^+U^*b = A^+b$ is the minimal least-squares solution of $Ax = b$. $\square$

1.2.3 Properties of Eigenvalues

In this subsection we state, in brief, some properties of eigenvalues and eigenvectors that we will use in the text. We also summarize the definitions of positive definite, negative definite and indefinite symmetric matrices and their characterizations in terms of eigenvalues.

The eigenvalue problem of a matrix A is that

$$Ax = \lambda x, \quad A \in R^{n \times n}, \quad x \neq 0, x \in R^n, \quad (1.2.56)$$

where λ is called an eigenvalue of A , x an eigenvector of A corresponding to λ , (λ, x) an eigen-pair of A .

The spectral radius of A is defined as

$$\rho(A) = \max_{1 \leq i \leq n} |\lambda_i|.$$

Let $A \in R^{m \times n}$ have singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n$, then

$$\begin{aligned} \|A\|_2 &= \sigma_1, \\ \|A\|_F^2 &= \sigma_1^2 + \cdots + \sigma_n^2. \end{aligned}$$

In particular, if $A \in R^{n \times n}$ is symmetric with eigenvalues $\lambda_1, \dots, \lambda_n$, then

$$\|A\|_2 = \max_{1 \leq i \leq n} |\lambda_i|.$$

Then we immediately have that if A is nonsingular, the condition number of A is

$$\kappa(A) = \frac{\sigma_1}{\sigma_n};$$

in addition, if A is symmetric, then

$$\kappa(A) = \frac{\max_i |\lambda_i|}{\min_i |\lambda_i|}.$$

Let $A \in R^{n \times n}$ with eigenvalues $\lambda_1, \dots, \lambda_n$. We have the following conclusions about the eigenvalues.

1. The eigenvectors corresponding to the distinct eigenvalues of A are independent.
2. A is diagonalizable if and only if, for each eigenvalue of A , its geometric multiplicity is equal to the algebraic multiplicity, i.e., the dimension of its corresponding eigenvectors is equal to the multiplicity of the eigenvalue.
3. Let $f(A)$ be a polynomial of A . If (λ, x) is an eigen-pair of A , then $(f(\lambda), x)$ is the eigen-pair of $f(A)$.

4. Let $B = PAP^{-1}$, where $P \in R^{n \times n}$ is a nonsingular transformation matrix. If (λ, x) is an eigen-pair of A , then (λ, Px) is the eigen-pair of B . This means that the similar transformation does not change the eigenvalues of a matrix.

Definition 1.2.8 *Let $A \in R^{n \times n}$ be symmetric. A is said to be positive definite if $v^T Av > 0, \forall v \in R^n, v \neq 0$. A is said to be positive semidefinite if $v^T Av \geq 0, \forall v \in R^n$. A is said to be negative definite or negative semidefinite if $-A$ is positive definite or positive semidefinite. A is said to be indefinite if it is neither positive semidefinite nor negative semidefinite.*

The main properties of a symmetric matrix are as follows. Let $A \in R^{n \times n}$ be symmetric. Then

- (1) All eigenvalues of A are real.
- (2) The eigenvectors corresponding to the distinct eigenvalues of A are orthogonal.
- (3) A is orthogonally similar to a diagonal matrix, i.e., there exists an $n \times n$ orthogonal matrix Q such that

$$Q^{-1}AQ = Q^T AQ = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix},$$

where $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A . This means a symmetric matrix has an orthonormal eigenvector system.

The following properties are about symmetric positive definite, symmetric positive semidefinite, and so on.

Let $A \in R^{n \times n}$ be symmetric. Then A is positive definite if and only if all its eigenvalues are positive. A is positive semidefinite if and only if all its eigenvalues are nonnegative. A is negative definite or negative semidefinite if and only if all its eigenvalues are negative or nonpositive. A is indefinite if and only if it has both positive and negative eigenvalues. Furthermore, A is positive definite if and only if A has a unique Cholesky factorization $A = LDL^T$ with all positive diagonal elements of D .

The following is the definition of the Rayleigh quotient of a matrix and its properties.

Definition 1.2.9 Let A be an $n \times n$ Hermitian matrix and $u \in C^n$. Then the Rayleigh quotient of A is defined by

$$R_\lambda(u) = \frac{u^* Au}{u^* u}, \quad u \neq 0. \quad (1.2.57)$$

Theorem 1.2.10 Let A be an $n \times n$ Hermitian matrix and $u \in C^n$. Then the Rayleigh quotient defined by (1.2.57) has the following basic properties:

(i) Homogeneous Property:

$$R_\lambda(\alpha u) = R_\lambda(u), \quad \alpha \neq 0. \quad (1.2.58)$$

(ii) Extreme Property:

$$\lambda_1 = \max_{\|u\|_2=1} u^* Au = \max_{u \neq 0} \frac{u^* Au}{u^* u}, \quad (1.2.59)$$

$$\lambda_n = \min_{\|u\|_2=1} u^* Au = \min_{u \neq 0} \frac{u^* Au}{u^* u}, \quad (1.2.60)$$

which show that the Rayleigh quotient has bounded property:

$$\lambda_n \leq R_\lambda(u) \leq \lambda_1. \quad (1.2.61)$$

(iii) Minimal Residual Property: for any $u \in C^n$,

$$\|(A - R_\lambda(u)I)u\| \leq \|(A - \mu I)u\|, \quad \forall \text{ real number } \mu. \quad (1.2.62)$$

Proof. Property (i) is immediate from Definition 1.2.9. Now we consider Property (ii). By Property (i), we can consider the Rayleigh quotient on a unit sphere, i.e.,

$$R_\lambda(u) = u^* Au, \quad \|u\|_2 = 1.$$

Let T be a unitary matrix such that $T^* AT = \Lambda$, where Λ is a diagonal matrix. Also let $u = Ty$, then

$$u^* Au = y^* \Lambda y = \sum_{i=1}^n \lambda_i |y_i|^2 \begin{cases} \geq \lambda_n \sum_{i=1}^n |y_i|^2, \\ \leq \lambda_1 \sum_{i=1}^n |y_i|^2. \end{cases}$$

Note that $\|u\|_2 = \|y\|_2 = 1$, hence the boundedness follows. Furthermore, when $y_1 = 1$ and $y_i = 0, i \neq 1$, λ_1 is the maximum; when $y_n = 1$ and $y_i = 0, i \neq n$, λ_n is the minimum. This proves Property (ii).

To establish Property (iii), we define

$$s(u) = Au - R_\lambda(u)u, \quad u \neq 0, \quad (1.2.63)$$

which implies that

$$Au = R_\lambda(u)u + s(u). \quad (1.2.64)$$

By Definition 1.2.9, we have $\langle s(u), u \rangle = \langle Au - R_\lambda(u)u, u \rangle = 0$ which means that the decomposition (1.2.64) is an orthogonal decomposition. Thus $R_\lambda(u)u$ is an orthogonal projection of Au on $L = \{u\}$, which shows that the residual defined by (1.2.63) has the minimal residual Property (iii). $\square$

Next, we state some concepts of reducible and irreducible matrices which are useful in discussing invertibility and positive definiteness of a matrix.

Definition 1.2.11 *Let $A \in R^{n \times n}$. A is said to be reducible if there is a permutation matrix P such that*

$$PAP^T = \begin{pmatrix} B_{11} & B_{12} \\ 0 & B_{22} \end{pmatrix},$$

where B_{11} and B_{22} are square matrices; A is irreducible if it is not reducible.

Equivalently, A is reducible if and only if there is a nonempty subset of indices $J \subset \{1, \dots, n\}$ such that

$$a_{kj} = 0, \quad \forall k \in J, j \notin J.$$

Definition 1.2.12 *Let $A \in R^{n \times n}$. A is said to be diagonally dominant if*

$$\sum_{j=1, j \neq i}^n |a_{ij}| \leq |a_{ii}|, \quad i = 1, \dots, n. \quad (1.2.65)$$

A is said to be strictly diagonally dominant if strict inequality holds in (1.2.65) for all i . A is said to be irreducibly diagonally dominant if it is irreducible, diagonally dominant, and strict inequality holds in (1.2.65) for at least one i .

The above concepts give an important theorem which is called the Diagonal Dominant Theorem.

Theorem 1.2.13 (*Diagonal Dominant Theorem*) *Let $A \in R^{n \times n}$ be either strictly or irreducibly diagonal dominant. Then A is invertible.*

As a corollary of the above theorem, we state the Gerschgorin circle Theorem which gives an isolation property of eigenvalues.

Theorem 1.2.14 *Let $A \in C^{n \times n}$. Define the i -th circle as*

$$\mathcal{D}_i = \{\lambda \mid |\lambda - a_{ii}| \leq \sum_{j=1, j \neq i}^n |a_{ij}|\}, \quad i = 1, \dots, n.$$

Then each eigenvalue of A lies in the union $S = \cup_{i=1}^n \mathcal{D}_i$. This also means that

$$\min_i \lambda_i \geq \min_i \{a_{ii} - \sum_{j=1, j \neq i}^n |a_{ij}|\}$$

and

$$\max_i \lambda_i \leq \max_i \{a_{ii} + \sum_{j=1, j \neq i}^n |a_{ij}|\}.$$

1.2.4 Rank-One Update

The rank-one update of matrices is often used in optimization. In this subsection we introduce inverse of rank-one update, determinant of rank-one update, chain of the eigenvalues of rank-one update, and updating matrix factorizations. Detailed proofs can be found in books on linear algebra or numerical linear algebra.

The following theorem due to Sherman and Morrison is wellknown.

Theorem 1.2.15 *Let $A \in R^{n \times n}$ be nonsingular and $u, v \in R^n$ be arbitrary. If*

$$1 + v^T A^{-1} u \neq 0, \tag{1.2.66}$$

then the rank-one update $A + uv^T$ of A is nonsingular, and its inverse is represented by

$$(A + uv^T)^{-1} = A^{-1} - \frac{A^{-1}uv^T A^{-1}}{1 + v^T A^{-1}u}. \tag{1.2.67}$$

An interesting generalization of the above theorem is

Theorem 1.2.16 *(Sherman-Morrison-Woodbury Theorem)*

Let A be an $n \times n$ nonsingular matrix, U, V $n \times m$ matrices. If $I + V^ A^{-1} U$ is invertible, then $A + UV^*$ is invertible, and*

$$(A + UV^*)^{-1} = A^{-1} - A^{-1}U(I + V^* A^{-1}U)^{-1}V^* A^{-1}. \tag{1.2.68}$$

Consider the determinant of a rank-one update; we have

$$\det(I + uv^T) = 1 + u^T v. \quad (1.2.69)$$

In fact, assuming $u \neq 0$, we have that the eigenvectors of $I + uv^T$ are either orthogonal to v or parallel to u . If they are orthogonal to v , the corresponding eigenvalues are 1; otherwise the corresponding eigenvalue is $1 + u^T v$. Hence (1.2.69) follows.

Furthermore, for the determinant of rank-two update, we have the following result:

$$\begin{aligned} & \det(I + u_1 u_2^T + u_3 u_4^T) \\ &= (1 + u_1^T u_2)(1 + u_3^T u_4) - (u_1^T u_4)(u_2^T u_3). \end{aligned} \quad (1.2.70)$$

In fact, as long as we note that

$$I + u_1 u_2^T + u_3 u_4^T = (I + u_1 u_2^T)[I + (I + u_1 u_2^T)^{-1} u_3 u_4^T],$$

it follows from (1.2.69) and (1.2.67) that

$$\begin{aligned} & \det(I + u_1 u_2^T + u_3 u_4^T) \\ &= (1 + u_1^T u_2)[1 + u_4^T (I + u_1 u_2^T)^{-1} u_3] \\ &= (1 + u_1^T u_2) \left[1 + u_4^T \left(I - \frac{u_1 u_2^T}{1 + u_1^T u_2} \right) u_3 \right] \\ &= (1 + u_1^T u_2)(1 + u_3^T u_4) - (u_1^T u_4)(u_2^T u_3). \end{aligned}$$

By $\|A\|_F^2 = \text{tr}(A^T A)$, where $\text{tr}(\cdot)$ denotes the trace of a matrix, it follows that the Frobenius norm of rank-one update $A + xy^T$ is

$$\|A + xy^T\|_F^2 = \|A\|_F^2 + 2y^T A^T x + \|x\|^2 \|y\|^2. \quad (1.2.71)$$

About the chain of the eigenvalues of rank-one update, we have the following theorem.

Theorem 1.2.17 *Let A be an $n \times n$ symmetric matrix with eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$. Also let $\bar{A} = A + \sigma u u^T$ with eigenvalues $\bar{\lambda}_1 \geq \bar{\lambda}_2 \geq \dots \geq \bar{\lambda}_n$, where $u \in R^n$. Then we have the conclusions:*

(i) *if $\sigma > 0$, then*

$$\bar{\lambda}_1 \geq \lambda_1 \geq \bar{\lambda}_2 \geq \lambda_2 \geq \dots \geq \bar{\lambda}_n \geq \lambda_n.$$

(ii) *if $\sigma < 0$, then*

$$\lambda_1 \geq \bar{\lambda}_1 \geq \lambda_2 \geq \bar{\lambda}_2 \geq \dots \geq \lambda_n \geq \bar{\lambda}_n.$$

Next, we discuss updating matrix factorizations which conclude updates of Cholesky factorization and orthogonal decomposition.

Let B and $\bar{B}$ be $n \times n$ symmetric and positive definite matrices,

$$\bar{B} = B + \alpha yy^T, \quad B = LDL^T. \quad (1.2.72)$$

We can find the Cholesky factorization $\bar{B} = \bar{L}\bar{D}\bar{L}^T$ as follows:

$$\begin{aligned} \bar{B} &= B + \alpha yy^T \\ &= L(D + \alpha pp^T)L^T, \end{aligned} \quad (1.2.73)$$

where p solves $Lp = y$. Note that since $D + \alpha pp^T$ is a positive definite matrix with the Cholesky factorization $D + \alpha pp^T = \hat{L}\hat{D}\hat{L}^T$, we have

$$\bar{B} = L\hat{L}\hat{D}\hat{L}^T L^T = \bar{L}\bar{D}\bar{L}^T, \quad (1.2.74)$$

where $\bar{L} = L\hat{L}$, $\bar{D} = \hat{D}$. The following algorithm gives the steps for computing $\bar{L}$ and $\bar{D}$.

Algorithm 1.2.18 (*Cholesky Factorization of Rank-One Update*)

1. Set $\alpha_1 = \alpha, w^{(1)} = y$.
2. For $j = 1, 2, \dots, n$, compute

$$\begin{aligned} p_j &= w_j^{(j)}, \\ \bar{d}_j &= d_j + \alpha_j p_j^2, \\ \beta_j &= p_j \alpha_j / \bar{d}_j, \\ \alpha_{j+1} &= d_j \alpha_j / \bar{d}_j, \\ w_r^{(j+1)} &= w_r^{(j)} - p_j l_{rj}, \quad r = j+1, \dots, n, \\ \bar{l}_{rj} &= l_{rj} + \beta_j w_r^{(j+1)}, \quad r = j+1, \dots, n. \quad \square \end{aligned}$$

Similarly, for the negative rank-one update of Cholesky factorization, we have

$$\begin{aligned} \bar{B} &= B - yy^T = L(D - pp^T)L^T \\ &= L\hat{L}\hat{D}\hat{L}^T L^T = \bar{L}\bar{D}\bar{L}^T. \end{aligned} \quad (1.2.75)$$

Since, in this case, it is possible that the elements of $\bar{D}$ become zero or negative due to round-off error, this phenomenon must be taken into consideration. The following algorithm keeps all $\bar{d}_j$ ($j = 1, \dots, n$) positive.

Algorithm 1.2.19 (*Cholesky Factorization of Negative Rank-One Update*)

1. Solve $Lp = y$ for p . Set $t_{n+1} = 1 - p^T D^{-1} p$. If $t_{n+1} < \epsilon_M$, set $t_{n+1} = \epsilon_M$, where ϵ_M is the relative precision of the computer.
2. For $j = n, n-1, \dots, 1$, compute

$$\begin{aligned}
 t_j &= t_{j+1} + p_j^2 / d_j, \\
 \bar{d}_j &= d_j t_{j+1} / t_j, \\
 \beta_j &= -p_j / (d_j t_{j+1}), \\
 w_j^{(j)} &= p_j, \\
 \bar{l}_{rj} &= l_{rj} + \beta_j w_r^{(j+1)}, \quad r = j+1, \dots, n., \\
 w_r^{(j)} &= w_r^{(j+1)} + p_j l_{rj}, \quad r = j+1, \dots, n. \quad \square
 \end{aligned}$$

Furthermore, Algorithm 1.2.18 and Algorithm 1.2.19 about Cholesky factorization of rank-one update can be used to compute the Cholesky factorization of rank-two update. Consider

$$\bar{B} = B + vw^T + wv^T. \quad (1.2.76)$$

Setting

$$x = (v + w) / \sqrt{2}, \quad y = (v - w) / \sqrt{2} \quad (1.2.77)$$

yields

$$\bar{B} = B + xx^T - yy^T, \quad (1.2.78)$$

so, we can use Algorithm 1.2.18 and Algorithm 1.2.19 to get the Cholesky factorization of $\bar{B}$.

Below, we consider the special cases of rank-two update. Let B be an $n \times n$ symmetric positive definite matrix with Cholesky factorization $B = LDL^T$. Consider the case adding one row and one column to B :

$$\bar{B} = \begin{bmatrix} B & b \\ b^T & \theta \end{bmatrix}, \quad (1.2.79)$$

where $b \in R^n$ and θ is a number. If we set

$$\hat{B} = \begin{bmatrix} B & 0 \\ 0 & \theta \end{bmatrix}, \quad (1.2.80)$$

then we have

$$\bar{B} = \hat{B} + e_{n+1} \begin{pmatrix} b \\ 0 \end{pmatrix}^T + \begin{pmatrix} b \\ 0 \end{pmatrix} e_{n+1}^T. \quad (1.2.81)$$

So, we can use the above algorithm to compute Cholesky factors $\bar{L}$ and $\bar{D}$ of $\bar{B}$. In addition, we can show that $\bar{L}$ and $\bar{D}$ have the following forms:

$$\bar{L} = \begin{bmatrix} L & 0 \\ l^T & 1 \end{bmatrix}, \quad \bar{D} = \begin{bmatrix} D & 0 \\ 0 & d \end{bmatrix}. \quad (1.2.82)$$

In fact, it is enough to consider

$$\begin{bmatrix} L & 0 \\ l^T & 1 \end{bmatrix} \begin{bmatrix} D & 0 \\ 0 & d \end{bmatrix} \begin{bmatrix} L^T & l \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} B & b \\ b^T & \theta \end{bmatrix} \quad (1.2.83)$$

and solve the equations obtained

$$\begin{aligned} LDL &= b, \\ d &= \theta - l^T D l \end{aligned} \quad (1.2.84)$$

for l and d . Then we get $\bar{L}$ and $\bar{D}$ from (1.2.82).

Now we consider the case deleting the j -th row and j -th column from B . Let $B = LDL^T$ with the form

$$B = \begin{bmatrix} B_1 & \vdots & B_2 \\ \cdots & \cdot & \cdots \\ B_2^T & \vdots & B_3 \end{bmatrix} \leftarrow j\text{-th row.} \quad (1.2.85)$$

Define

$$\bar{B} = \begin{bmatrix} B_1 & B_2 \\ B_2^T & B_3 \end{bmatrix} \} n-1 \text{ columns.} \quad (1.2.86)$$

The algebraic operations give

$$\bar{B} = \hat{L} D \hat{L}^T, \quad (1.2.87)$$

which is our desired result, where $\hat{L}$ is an $(n-1) \times n$ matrix obtained by deleting the j -th row from L .

In the above, we discussed the Cholesky factorization of rank-one update. Next, we handle the QR factorization of rank-one update. Let $A, \bar{A} \in R^{n \times n}$, $u, v \in R^n$,

$$A = QR, \quad \bar{A} = A + uv^T. \quad (1.2.88)$$

Then we have

$$\bar{A} = QR + uv^T = Q(R + wv^T), \quad (1.2.89)$$

where $w = Q^T u$. Forming QR decomposition

$$R + wv^T = \tilde{Q}\tilde{R},$$

we have

$$\bar{A} = Q\tilde{Q}\tilde{R} \triangleq \bar{Q}\bar{R}, \quad (1.2.90)$$

where $\bar{Q} = Q\tilde{Q}$, $\bar{R} = \tilde{R}$.

Similarly, if $m \times n$ matrix A ($m < n$) has an orthogonal decomposition

$$A = [L \ 0]Q, \quad (1.2.91)$$

where L is an $m \times m$ unit lower triangular matrix and Q is an $n \times n$ orthogonal matrix with $Q^T Q = I$, then we can obtain the LQ decomposition of

$$\bar{A} = A + xy^T \quad (1.2.92)$$

as follows.

$$\begin{aligned} \bar{A} &= A + xy^T \\ &= [L \ 0]Q + xy^T \\ &= ([L \ 0] + xw^T)Q \quad (\text{where } w = Qy) \\ &= ([L \ 0] + xw^T)P^T P Q \quad (\text{where } P^T P = I) \\ &= ([H \ 0] + \alpha x e_1^T)P Q \quad (\text{where } Pw = \alpha e_1, H = LP^T) \\ &= [\bar{H} \ 0]P Q \\ &= [\bar{H} \ 0]\bar{P}\bar{P}^T P Q \quad (\text{where } \bar{P}\bar{P}^T = I) \\ &= [\bar{L} \ 0]\bar{P}^T P Q \quad (\text{where } [\bar{H} \ 0]\bar{P} = [\bar{L} \ 0]) \\ &= [\bar{L} \ 0]\bar{Q} \quad (\text{where } \bar{Q} = \bar{P}^T P Q). \end{aligned} \quad (1.2.93)$$

1.2.5 Function and Differential

This subsection presents some materials of set theory and multivariable calculus background.

Give a point $x \in R^n$ and a $\delta > 0$. The δ -neighborhood of x is defined as

$$N_\delta(x) = \{y \in R^n \mid \|y - x\| < \delta\}.$$

Let $D \subset R^n$ and $x \in D$. The point x is said to be an interior point of D if there exists a δ -neighborhood of x such that $N_\delta(x) \subset D$. The set of all such points is called the interior of D and is denoted by $\text{int}(D)$. Obviously, $\text{int}(D) \subset D$. Furthermore, if $\text{int}(D) = D$, i.e., every point of D is the interior point of D , then D is an open set.

$x \in D \subset R^n$ is said to be an accumulation point if for each $\delta > 0$, $D \cap N_\delta(x) \neq \phi$, where ϕ is an empty set. It means that there exists a subsequence $\{x_{n_k}\} \subset D$, such that $x_{n_k} \rightarrow x$. The set of all such points is called the closure of D and is denoted by $\bar{D}$. Obviously, $D \subset \bar{D}$. Furthermore, if $D = \bar{D}$, i.e., every accumulation point of D is contained in D , then D is said to be closed. It is also clear that a set $D \subset R^n$ is closed if and only if its complement is open.

A set $D \subset R^n$ is said to be compact if it is bounded and closed. For every sequence $\{x_k\}$ in a compact set D , there exists a convergent subsequence with a limit in D .

A function $f : R^n \rightarrow R$ is said to be continuous at $\bar{x} \in R^n$ if, for any given $\epsilon > 0$, there exists $\delta > 0$ such that $\|x - \bar{x}\| < \delta$ implies $|f(x) - f(\bar{x})| < \epsilon$. It can also be written as follows: $\forall \epsilon > 0, \exists \delta > 0$, such that $\forall x \in N_\delta(\bar{x})$, we have $f(x) \in N_\epsilon(f(\bar{x}))$. If f is continuous at every point in an open set $D \subset R^n$, then f is said to be continuous on D .

A continuous function $f : R^n \rightarrow R$ is said to be continuously differentiable at $x \in R^n$, if $\left(\frac{\partial f}{\partial x_i}\right)(x)$ exists and is continuous, $i = 1, \dots, n$. The gradient of f at x is defined as

$$\nabla f(x) = \left[\frac{\partial f}{\partial x_1}(x), \dots, \frac{\partial f}{\partial x_n}(x) \right]^T. \quad (1.2.94)$$

If f is continuously differentiable at every point of an open set $D \subset R^n$, then f is said to be continuously differentiable on D and denoted by $f \in C^1(D)$.

A continuously differentiable function $f : R^n \rightarrow R$ is called twice continuously differentiable at $x \in R^n$ if $\frac{\partial^2 f}{\partial x_i \partial x_j}(x)$ exists and is continuous, $i = 1, \dots, n$. The Hessian of f is defined as the $n \times n$ symmetric matrix with elements

$$[\nabla^2 f(x)]_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}(x), \quad 1 \leq i, j \leq n.$$

If f is twice continuously differentiable at every point in an open set $D \subset R^n$, then f is said to be twice continuously differentiable on D and denoted by $f \in C^{(2)}(D)$.

Let $f : R^n \rightarrow R$ be continuously differentiable on an open set $D \subset R^n$. Then for $x \in D$ and $d \in R^n$, the directional derivative of f at x in the direction d is defined as

$$f'(x; d) \stackrel{\text{def}}{=} \lim_{\theta \rightarrow 0} \frac{f(x + \theta d) - f(x)}{\theta} = \nabla f(x)^T d, \quad (1.2.95)$$

where $\nabla f(x)$ is the gradient of f at x , an $n \times 1$ vector.

For any $x, x + d \in D$, if $f \in C^1(D)$, then

$$\begin{aligned} f(x + d) &= f(x) + \int_0^1 \nabla f(x + td)^T d dt \\ &= f(x) + \int_x^{x+d} \nabla f(\xi) d\xi. \end{aligned} \quad (1.2.96)$$

Thus,

$$f(x + d) = f(x) + \nabla f(\xi)^T d, \quad \xi \in (x, x + d). \quad (1.2.97)$$

Similarly, for all $x, y \in D$, we have

$$f(y) = f(x) + \nabla f(x + t(y - x))^T (y - x), \quad t \in (0, 1), \quad (1.2.98)$$

or

$$f(y) = f(x) + \nabla f(x)^T (y - x) + o(\|y - x\|). \quad (1.2.99)$$

It follows from (1.2.98) that

$$|f(y) - f(x)| \leq \|y - x\| \sup_{\xi \in L(x, y)} \|\nabla f(\xi)\|, \quad (1.2.100)$$

where $L(x, y)$ denotes the line segment with endpoints x and y .

Let $f \in C^{(2)}(D)$. For any $x \in D, d \in R^n$, the second directional derivative of f at x in the direction d is defined as

$$f''(x; d) = \lim_{\theta \rightarrow 0} \frac{f'(x + \theta d; d) - f'(x; d)}{\theta}, \quad (1.2.101)$$

which equals $d^T \nabla^2 f(x) d$, where $\nabla^2 f(x)$ denotes the Hessian of f at x . For any $x, x + d \in D$, there exists $\xi \in (x, x + d)$ such that

$$f(x + d) = f(x) + \nabla f(x)^T d + \frac{1}{2} d^T \nabla^2 f(\xi) d, \quad (1.2.102)$$

or

$$f(x + d) = f(x) + \nabla f(x)^T d + \frac{1}{2} d^T \nabla^2 f(x) d + o(\|d\|^2). \quad (1.2.103)$$

Let $h : R^n \rightarrow R$, $g : R^m \rightarrow R$, $f : R^n \rightarrow R^m$. Let $f \in C^1$, $g \in C^1$, $h(x_0) = g(f(x_0))$. Then the chain rule is

$$h'(x_0) = g'(f(x_0))f'(x_0), \quad (1.2.104)$$

where $f'(x_0) = \left[\frac{\partial f_i(x_0)}{\partial x^j} \right]_{m \times n}$ is an $m \times n$ matrix. Also

$$h''(x_0) = \nabla f(x_0)^T \nabla^2 g[f(x_0)] \nabla f(x_0) + \sum_{i=1}^m \frac{\partial g[f(x_0)]}{\partial f_i} [f_i(x_0)]''. \quad (1.2.105)$$

Next, we discuss the calculus of vector-valued functions.

A continuous function $F : R^n \rightarrow R^m$ is continuously differentiable at $x \in R^n$ if each component function $f_i (i = 1, \dots, m)$ is continuously differentiable at x . The derivative $F'(x) \in R^{m \times n}$ of F at x is called the Jacobian matrix of F at x ,

$$F'(x) = J(x)$$

with components

$$[F'(x)]_{ij} = [J(x)]_{ij} = \frac{\partial f_i}{\partial x^j}(x), \quad i = 1, \dots, m; j = 1, \dots, n.$$

If $F : R^n \rightarrow R^m$ is continuously differentiable in an open convex set $D \subset R^n$, then for any $x, x + d \in D$, we have

$$F(x + d) - F(x) = \int_0^1 J(x + td) dtdt = \int_x^{x+d} F'(\xi) d\xi. \quad (1.2.106)$$

In many of our considerations, we shall wish to single out different types of continuities.

Definition 1.2.20 $F : D \subset R^n \rightarrow R^m$ is Hölder continuous on D if there exist constants $\gamma \geq 0$ and $p \in (0, 1]$ so that for all $x, y \in D$,

$$\|F(y) - F(x)\| \leq \gamma \|y - x\|^p. \quad (1.2.107)$$

If $p = 1$, then F is called Lipschitz continuous on D and γ is a Lipschitz constant.

$F : D \subset R^n \rightarrow R^m$ is Hölder continuous at $x \in D$ if (1.2.107) holds for any y in the neighborhood of x .

Definition 1.2.21 $F : D \subset R^n \rightarrow R^m$ is *hemi-continuous* at $x \in D$ if, for any $d \in R^n$ and $\epsilon > 0$, there is a $\delta = \delta(\epsilon, d)$ so that whenever $|t| < \delta$ and $x + td \in D$,

$$\|F(x + td) - F(x)\| < \epsilon. \quad (1.2.108)$$

We also can define the upper hemi-continuous and lower hemi-continuous at $x \in D$ if, instead of (1.2.108), we use, respectively, $F(x + td) < F(x) + \epsilon$ and $F(x + td) > F(x) - \epsilon$ for sufficiently small t .

The following two theorems establish the bounds of errors within which some standard models approximate the objective functions. For $F : R^n \rightarrow R^m$, Theorem 1.2.22 gives a bound of the error in linear model $F(x) + F'(x)d$ as an approximation to $F(x + d)$. Similarly, for $f : R^n \rightarrow R$, Theorem 1.2.23 gives a bound of errors with a quadratic model as an approximation to $f(x + d)$.

Theorem 1.2.22 Let $F : R^n \rightarrow R^m$ be continuously differentiable in the open convex set $D \subset R^n$. Let F' be Lipschitz continuous at $x \in D$. Then for any $x + d \in D$, we have

$$\|F(x + d) - F(x) - F'(x)d\| \leq \frac{\gamma}{2}\|d\|^2. \quad (1.2.109)$$

Proof.

$$\begin{aligned} F(x + d) - F(x) - F'(x)d &= \int_0^1 F'(x + \alpha d) d \alpha - F'(x)d \\ &= \int_0^1 [F'(x + \alpha d) - F'(x)] d \alpha. \end{aligned}$$

Hence

$$\begin{aligned} \|F(x + d) - F(x) - F'(x)d\| &\leq \int_0^1 \|F'(x + \alpha d) - F'(x)\| \|d\| d\alpha \\ &\leq \int_0^1 \gamma \|\alpha d\| \|d\| d\alpha \\ &= \gamma \|d\|^2 \int_0^1 \alpha d\alpha \\ &= \frac{\gamma}{2} \|d\|^2. \quad \square \end{aligned}$$

Theorem 1.2.23 *Let $f : R^n \rightarrow R$ be twice continuously differentiable in the open convex set $D \subset R^n$. Let $\nabla^2 f(x)$ be Lipschitz continuous at $x \in D$ with Lipschitz constant γ . Then for any $x + d \in D$, we have*

$$\left| f(x + d) - [f(x) + \nabla f(x)^T d + \frac{1}{2} d^T \nabla^2 f(x) d] \right| \leq \frac{\gamma}{6} \|d\|^3. \quad (1.2.110)$$

The proof of this theorem is left to readers as an exercise.

As a generalization of Theorem 1.2.22, we obtain

Theorem 1.2.24 *Let $F : R^n \rightarrow R^m$ be continuously differentiable in the open convex set $D \subset R^n$. Then for any $u, v, x \in D$, we have*

$$\begin{aligned} & \|F(u) - F(v) - F'(x)(u - v)\| \\ & \leq \left[\sup_{0 \leq t \leq 1} \|F'(v + t(u - v)) - F'(x)\| \right] \|u - v\|. \end{aligned} \quad (1.2.111)$$

Furthermore, assume that F' is Lipschitz continuous in D , then

$$\|F(u) - F(v) - F'(x)(u - v)\| \leq \gamma \sigma(u, v) \|u - v\| \quad (1.2.112)$$

and

$$\|F(u) - F(v) - F'(x)(u - v)\| \leq \gamma \frac{\|u - x\| + \|v - x\|}{2} \|u - v\|, \quad (1.2.113)$$

where $\sigma(u, v) = \max\{\|u - x\|, \|v - x\|\}$.

Proof. By (1.2.106) and the mean-value theorem of integration, we have

$$\begin{aligned} & \|F(u) - F(v) - F'(x)(u - v)\| \\ & = \left\| \int_0^1 [F'(v + t(u - v)) - F'(x)](u - v) dt \right\| \\ & \leq \int_0^1 \|F'(v + t(u - v)) - F'(x)\| \|u - v\| dt \\ & \leq \left[\sup_{0 \leq t \leq 1} \|F'(v + t(u - v)) - F'(x)\| \right] \|u - v\| \end{aligned}$$

which is (1.2.111). Also since F' is Lipschitz continuous in D , we proceed with the above inequality and get

$$\|F(u) - F(v) - F'(x)(u - v)\|$$

$$\begin{aligned}
&\leq \gamma \int_0^1 \|v + t(u - v) - x\| \|u - v\| dt \\
&\leq \gamma \sup_{0 \leq t \leq 1} \|v + t(u - v) - x\| \|u - v\| \\
&= \gamma \sigma(u, v) \|u - v\|
\end{aligned}$$

which is (1.2.112). Similarly, we can derive (1.2.113) which is left as an exercise. $\square$

The following theorem is useful, giving a relation between $\|F(u) - F(v)\|$ and $\|u - v\|$.

Theorem 1.2.25 *Let F and F' satisfy the conditions of Theorem 1.2.24. Assume that $[F'(x)]^{-1}$ exists. Then there exist $\epsilon > 0$ and $\beta > \alpha > 0$ such that for all $u, v \in D$, when $\max\{\|u - x\|, \|v - x\|\} \leq \epsilon$, we have*

$$\alpha \|u - v\| \leq \|F(u) - F(v)\| \leq \beta \|u - v\|. \quad (1.2.114)$$

Proof. By the triangle inequality and (1.2.112),

$$\begin{aligned}
\|F(u) - F(v)\| &\leq \|F'(x)(u - v)\| + \|F(u) - F(v) - F'(x)(u - v)\| \\
&\leq (\|F'(x)\| + \gamma \sigma(u, v)) \|u - v\| \\
&\leq (\|F'(x)\| + \gamma \epsilon) \|u - v\|.
\end{aligned}$$

Set $\beta = \|F'(x)\| + \gamma \epsilon$, we obtain the right inequality of (1.2.114). Similarly,

$$\begin{aligned}
\|F(u) - F(v)\| &\geq \|F'(x)(u - v)\| - \|F(u) - F(v) - F'(x)(u - v)\| \\
&\geq \left[1 / \|[F'(x)]^{-1}\| - \gamma \sigma(u, v)\right] \|u - v\| \\
&\geq [1 / \|[F'(x)]^{-1}\| - \gamma \epsilon] \|u - v\|.
\end{aligned}$$

Hence, if $\frac{1}{\|[F'(x)]^{-1}\|} > \gamma \epsilon$, the left inequality of (1.2.114) also holds with

$$\alpha = \frac{1}{\|[F'(x)]^{-1}\|} - \gamma \epsilon > 0. \quad \square$$

Corollary 1.2.26 *Let F and F' satisfy the conditions of Theorem 1.2.22. When u and v are sufficiently close to x , we have*

$$\limsup_{\omega \rightarrow 0} \frac{\|u - x\|}{\|v - x\|} \leq C \limsup_{\omega \rightarrow 0} \frac{\|F(u) - F(x)\|}{\|F(v) - F(x)\|}, \quad (1.2.115)$$

where $C = \|F'(x)\| \|[F'(x)]^{-1}\|$ is a constant and $\omega = \max\{\|u - x\|, \|v - x\|\}$.

Proof. By using Theorem 1.2.22, we have

$$\begin{aligned}\|F(v) - F(x)\| &\leq \|F'(x)(v - x)\| + \|F(v) - F(x) - F'(x)(v - x)\| \\ &\leq \|F'(x)\|\|v - x\| + O(\|v - x\|^2)\end{aligned}$$

and

$$\begin{aligned}\|F(u) - F(x)\| &\geq \|F'(x)(u - x)\| - \|F(u) - F(x) - F'(x)(u - x)\| \\ &\geq \|u - x\|/\|[F'(x)]^{-1}\| + O(\|u - x\|^2).\end{aligned}$$

Then

$$\frac{\|F(u) - F(x)\|}{\|F(v) - F(x)\|} \geq \frac{\|u - x\|/\|F'(x)^{-1}\| + O(\|u - x\|^2)}{\|F'(x)\|\|v - x\| + O(\|v - x\|^2)}.$$

Setting $C = \|F'(x)\|\|F'(x)^{-1}\|$ and taking limit give

$$C \limsup_{\omega \rightarrow 0} \frac{\|F(u) - F(x)\|}{\|F(v) - F(x)\|} \geq \limsup_{\omega \rightarrow 0} \frac{\|u - x\|}{\|v - x\|},$$

where $\omega = \max\{\|u - x\|, \|v - x\|\}$. $\square$

We conclude this subsection with some remarks about differentiation of the vector-valued functions.

About the calculus of vector-valued functions, we would like to review Gateaux and Fréchet derivatives.

Definition 1.2.27 Let $D \subset R^n$ be an open set. The function $F : D \subset R^n \rightarrow R^m$ is Gateaux- (or G -) differentiable at $x \in D$ if there exists a linear operator $A \in L(R^n, R^m)$ such that for any $d \in R^n$,

$$\lim_{\alpha \rightarrow 0} \frac{1}{\alpha} \|F(x + \alpha d) - F(x) - \alpha Ad\| = 0. \quad (1.2.116)$$

The linear operator A is denoted by $F'(x)$ and is called the G -derivative of F at x .

Definition 1.2.28 Let $D \subset R^n$ be an open set. The function $F : R^n \rightarrow R^m$ is Fréchet- (or F -) differentiable at $x \in D$ if there is a linear operator $A \in L(R^n, R^m)$ such that for any $d \in R^n$,

$$\lim_{d \rightarrow 0} \frac{\|F(x + d) - F(x) - Ad\|}{\|d\|} = 0. \quad (1.2.117)$$

The linear operator A is again denoted by $F'(x)$, and is called the F -derivative of F at x .

The F -differentiability can also be written as

$$F(x + d) - F(x) = F'(x)d + o(\|d\|).$$

Furthermore, if for any $u, v \in R^n$,

$$\lim_{\|u-v\| \rightarrow 0} \frac{\|F(u) - F(v) - F'(x)(u - v)\|}{\|u - v\|} = 0, \quad (1.2.118)$$

then F is called strongly F -differentiable at $x \in D$ and $F'(x)$ is called a strong F -derivative.

From the above two definitions, we know the following facts.

1. If $F : R^n \rightarrow R^m$ is continuous at $x \in R^n$, then F is hemi-continuous at x .
2. If $F : R^n \rightarrow R^m$ is G-differentiable at $x \in D$, then F is hemi-continuous at x .
3. If $F : R^n \rightarrow R^m$ is F-differentiable at $x \in D$, then F is continuous at x .
4. If F is F-differentiable at $x \in D$, then it is G-differentiable at x ; however, the reverse is not true.
5. If F is G-differentiable and its G-derivative F' is continuous, then F is F-differentiable and the F-derivative is continuous. In this case, we say that F is continuously differentiable.
6. The G-derivative and F-derivative of F , if they exist, are equal and given by the Jacobian matrix

$$F'(x) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1}(x) & \frac{\partial f_1}{\partial x_2}(x) & \cdots & \frac{\partial f_1}{\partial x_n}(x) \\ \cdots & \cdots & \cdots & \cdots \\ \frac{\partial f_m}{\partial x_1}(x) & \frac{\partial f_m}{\partial x_2}(x) & \cdots & \frac{\partial f_m}{\partial x_n}(x) \end{bmatrix},$$

where $f_1, f_2, \dots, f_m$ are components of F .

7. The mean-value theorem: Let $F : R^n \rightarrow R^m$ be G-differentiable in the open convex set $D \subset R^n$. Then we have the following forms of the mean-value theorem:

(a) For any $x, y, z \in D$, there exist $t_1, t_2, \dots, t_m \in [0, 1]$ such that

$$F(y) - F(x) = \begin{pmatrix} f'_1(x + t_1(y - x)) \\ f'_2(x + t_2(y - x)) \\ \vdots \\ f'_m(x + t_m(y - x)) \end{pmatrix} \quad (1.2.119)$$

and

$$\|F(y) - F(x)\| \leq \sup_{0 \leq t \leq 1} \|F'(x + t(y - x))\| \|y - x\|.$$

(b) For any $x, y, z \in D$,

$$\|F(y) - F(z) - F'(x)(y - z)\| \leq \sup_{0 \leq t \leq 1} \|F'(z + t(y - z)) - F'(x)\| \|y - z\|. \quad (1.2.120)$$

(c) Furthermore, if the G-derivative F' is hemi-continuous on D , then for any $x, y \in D$,

$$F(y) - F(x) = \int_0^1 F'(x + t(y - x))(y - x) dt. \quad (1.2.121)$$

(d) If assume also that $F'(x)$ is Hölder continuous on D , then for all $x, y \in D$,

$$\|F(y) - F(x) - F'(x)(y - x)\| \leq \frac{\gamma}{p+1} \|y - x\|^{p+1}. \quad (1.2.122)$$

If $p = 1$, it is just (1.2.109).

1.3 Convex Sets and Convex Functions

Convex sets and convex functions play an important role in the study of optimization. In this section, we introduce the fundamental concepts and results of convex sets and convex functions.

1.3.1 Convex Sets

Definition 1.3.1 Let the set $S \subset R^n$. If, for any $x_1, x_2 \in S$, we have

$$\alpha x_1 + (1 - \alpha)x_2 \in S, \forall \alpha \in [0, 1], \quad (1.3.1)$$

then S is said to be a convex set.

This definition indicates, in geometry, that for any two points $x_1, x_2 \in S$, the line segment joining x_1 and x_2 is entirely contained in S . It also states that S is path-connected, i.e., two arbitrary points in S can be linked by a continuous path.

It can be shown by induction that the set $S \subset R^n$ is convex if and only if for any $x_1, x_2, \dots, x_m \in S$,

$$\sum_{i=1}^m \alpha_i x_i \in S, \quad (1.3.2)$$

where $\sum_{i=1}^m \alpha_i = 1, \alpha_i \geq 0, i = 1, \dots, m$.

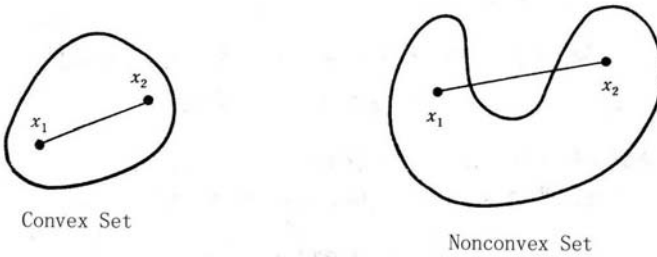


Figure 1.3.1 Convex set and nonconvex set

In (1.3.1), $x = \alpha x_1 + (1 - \alpha)x_2$, where $\alpha \in [0, 1]$, is called a convex combination of x_1 and x_2 . In (1.3.2), $x = \sum_{i=1}^m \alpha_i x_i$ is called a convex combination of $x_1, \dots, x_m$, where $\sum_{i=1}^m \alpha_i = 1, \alpha_i \geq 0, i = 1, \dots, m$.

Example 1.3.2 The hyperplane $H = \{x \in R^n \mid p^T x = \alpha\}$ is a convex set, where $p \in R^n$ is a nonzero vector referred to as the normal vector to the hyperplane, and α is a scalar.

In fact, for any $x_1, x_2 \in H$ and each $\theta \in [0, 1]$,

$$p^T[\theta x_1 + (1 - \theta)x_2] = \alpha,$$

then $\theta x_1 + (1 - \theta)x_2 \in H$.

In the hyperplane $H = \{x \in R^n \mid p^T x = \alpha\}$, if $\alpha = 0$, it can be reduced to a subspace of vectors that are orthogonal to p .

Similarly, the closed half space $H^- = \{x \in R^n \mid p^T x \leq \beta\}$ and $H^+ = \{x \in R^n \mid p^T x \geq \beta\}$ are closed convex sets. The open half space $(\overset{\circ}{H})^- = \{x \in R^n \mid p^T x < \beta\}$ and $(\overset{\circ}{H})^+ = \{x \in R^n \mid p^T x > \beta\}$ are open convex sets.

Example 1.3.3 The ray $S = \{x \in R^n \mid x = x_0 + \lambda d, \lambda \geq 0\}$ is a convex set, where $d \in R^n$ is a nonzero vector, and $x_0 \in R^n$ is a fixed point.

In fact, for any $x_1, x_2 \in S$ and each $\lambda \in [0, 1]$, we have

$$x_1 = x_0 + \lambda_1 d, \quad x_2 = x_0 + \lambda_2 d,$$

where $\lambda_1, \lambda_2 \in [0, 1]$. Hence

$$\begin{aligned} \lambda x_1 + (1 - \lambda)x_2 &= \lambda(x_0 + \lambda_1 d) + (1 - \lambda)(x_0 + \lambda_2 d) \\ &= x_0 + [\lambda\lambda_1 + (1 - \lambda)\lambda_2]d. \end{aligned}$$

Since $\lambda\lambda_1 + (1 - \lambda)\lambda_2 \geq 0$, then $\lambda x_1 + (1 - \lambda)x_2 \in S$.

The finite intersection of closed half spaces

$$S = \{x \in R^n \mid p_i^T x \leq \beta_i, \quad i = 1, \dots, m\},$$

is called a polyhedral set, where p_i is a nonzero vector, β_i a scalar. The polyhedral is a convex set.

Since an equality can be represented by two inequalities, the following sets are examples of polyhedral sets:

$$S = \{x \in R^n \mid Ax = b, x \geq 0\},$$

$$S = \{x \in R^n \mid Ax \geq 0, x \geq 0\}.$$

The theorems below state the algebraic properties and topological properties. That is the intersection of two convex sets is convex, the algebraic sum of two convex sets is convex, the interior of a convex set is convex, and the closure of a convex set is convex.

Theorem 1.3.4 *Let S_1 and S_2 be convex sets in R^n . Then*

1. $S_1 \cap S_2$ is convex;
2. $S_1 \pm S_2 = \{x_1 \pm x_2 \mid x_1 \in S_1, x_2 \in S_2\}$ is convex.

Proof. The proof is immediate from the definition of convex set and left to readers as an exercise. $\square$

From this theorem, we know that the feasible regions in linear programming and quadratic programming are convex sets, because they are the intersection of a hyperplane and a half space.

Theorem 1.3.5 *Let $S \subset R^n$ be a convex set. Then*

1. the interior $\text{int}S$ of S is a convex set;
2. the closure $\bar{S}$ of S is a convex set.

Proof. 1) Let x and x' be in $\text{int}S$, and $x'' = \alpha x + (1 - \alpha)x'$, $\alpha \in (0, 1)$. Choose $\delta > 0$ such that $B(x', \delta) \subset S$, where $B(x', \delta)$ is the δ -neighborhood of x' . It is easy to see that $\|x'' - x\|/\|x' - x\| = 1 - \alpha$. We know that $B(x'', (1 - \alpha)\delta)$ is just the set $\alpha x + (1 - \alpha)B(x', \delta)$ which is in S . Therefore $B(x'', (1 - \alpha)\delta) \subset S$ which shows that $x'' \in \text{int} S$.

2) Take $x, x' \in \bar{S}$. Select in S two sequences $\{x_k\}$ and $\{x'_k\}$ converging to x and x' respectively. Then, for $\alpha \in [0, 1]$, we have

$$\begin{aligned} & \|[\alpha x_k + (1 - \alpha)x'_k] - [\alpha x + (1 - \alpha)x']\| \\ &= \|\alpha(x_k - x) + (1 - \alpha)(x'_k - x')\| \\ &\leq \alpha\|x_k - x\| + (1 - \alpha)\|x'_k - x'\|. \end{aligned}$$

Taking the limit yields

$$\lim_{k \rightarrow \infty} \|[\alpha x_k + (1 - \alpha)x'_k] - [\alpha x + (1 - \alpha)x']\| = 0,$$

which shows $\alpha x + (1 - \alpha)x' \in \bar{S}$. $\square$

Now we state some concepts related to convex sets.

Let $S \subset R^n$ be a nonempty set. We define the convex hull $\text{conv}(S)$ as the intersection of all convex sets containing S , which is described as the set of

all convex combinations of the elements of S :

$$\begin{aligned}\text{conv}(S) &\triangleq \cap \{C \mid C \text{ is convex and contains } S\} \\ &= \{x \in R^n \mid x = \sum_{i=1}^m \alpha_i x_i, x_i \in S, \sum_{i=1}^m \alpha_i = 1, \\ &\quad \alpha_i \geq 0, i = 1, \dots, m\}.\end{aligned}\tag{1.3.3}$$

We can see that $\text{conv}(S)$ is the smallest convex set containing S .

A nonempty set $C \subset R^n$ is called a cone if it is closed under positive scalar multiplication, i.e., if $x \in C$ implies that $\lambda x \in C$ for all $\lambda > 0$. If, in addition, C is convex, then C is called a convex cone. $C \subset R^n$ is a convex cone if and only if it is closed under addition and positive scalar multiplication. The smallest convex cone containing convex S is

$$C = \{\lambda x \mid \lambda > 0, x \in S\}.$$

The following are examples of convex cones. For example, the nonnegative orthant of R^n

$$\{x = (\xi_1, \dots, \xi_n) \mid \xi_1 \geq 0, \dots, \xi_n \geq 0\},$$

positive orthant of R^n

$$\{x = (\xi_1, \dots, \xi_n) \mid \xi_1 > 0, \dots, \xi_n > 0\}$$

and the intersection of m half-spaces

$$\{x \in R^n \mid x^T b_i \leq 0, b_i \in R^n, i = 1, \dots, m\}$$

are convex cones .

A specially important class of convex cones is polar cone. Let S be a nonempty set in R^n . The polar cone of S , denoted by S^* , is given by $\{p \mid p^T x \leq 0 \text{ for all } x \in S\}$. It is easy to see from the above definition that the polar cone S^* of a nonempty set S has the following properties:

1. S^* is a closed convex cone.
2. $S \subset S^{**}$, where S^{**} is the polar cone of S^* . If S is a nonempty closed convex set, then $S^{**} = S$.
3. If S_1, S_2 are nonempty sets, then $S_1 \subset S_2$ implies $S_2^* \subset S_1^*$.

The normal and tangent cones play a special role in constrained optimization. Here we give their definitions below. Let S be a closed convex set. The normal cone of S at $\bar{x}$ is defined as

$$N(\bar{x}) = \{y \in R^n \mid \langle y, x - \bar{x} \rangle \leq 0, \forall x \in S\}. \quad (1.3.4)$$

The tangent cone of S at $\bar{x} \in S$ is the polar of the normal cone at $\bar{x}$, that is

$$\begin{aligned} T(\bar{x}) &= (N(\bar{x}))^* = \text{cl}\{\lambda(x - \bar{x}) \mid \lambda \geq 0, x \in S\} \\ &= \{d \mid d = \lim_{x \rightarrow \bar{x}} \lambda(x - \bar{x}), \lambda \geq 0, x \in S\}, \end{aligned} \quad (1.3.5)$$

where $\text{cl}\{S\}$ denotes the closure of S .

1.3.2 Convex Functions

Definition 1.3.6 Let $S \subset R^n$ be a nonempty convex set. Let $f : S \subset R^n \rightarrow R$. If, for any $x_1, x_2 \in S$ and all $\alpha \in (0, 1)$, we have

$$f(\alpha x_1 + (1 - \alpha)x_2) \leq \alpha f(x_1) + (1 - \alpha)f(x_2), \quad (1.3.6)$$

then f is said to be convex on S . If the above inequality is true as a strict inequality for all $x_1 \neq x_2$, i.e.,

$$f(\alpha x_1 + (1 - \alpha)x_2) < \alpha f(x_1) + (1 - \alpha)f(x_2), \quad (1.3.7)$$

then f is called a strict convex function on S . If there is a constant $c > 0$ such that for any $x_1, x_2 \in S$,

$$f(\alpha x_1 + (1 - \alpha)x_2) \leq \alpha f(x_1) + (1 - \alpha)f(x_2) - \frac{1}{2}c\alpha(1 - \alpha)\|x_1 - x_2\|^2, \quad (1.3.8)$$

then f is called a uniformly (or strongly) convex function on S .

If $-f$ is a convex (strictly convex, uniformly convex) function on S , then f is said to be a concave (strictly concave, uniformly concave) function.

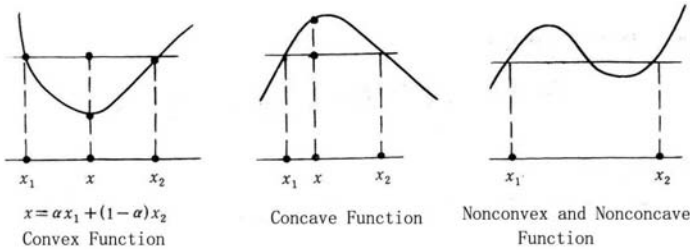


Figure 1.3.2 Convex function and concave function

Figure 1.3.2 gives examples of convex, concave, and neither convex nor concave functions. The geometrical interpretation of a convex function says that the function values are below the corresponding chord, that is, the values of a convex function at points on the line segment $\alpha x_1 + (1 - \alpha)x_2$ are less than or equal to the height of the chord joining the points $(x_1, f(x_1))$ and $(x_2, f(x_2))$. It is obvious from the definition of convex function that a linear function $f(x) = a^T x + \beta$ is both a convex and concave function on R^n , where $a, x \in R^n, \beta \in R$.

The other basic and important examples of convex functions are indicator function, support function, norm and distance function.

Let $S \subset R^n$ be a nonempty subset; the indicator function $I_S : R^n \rightarrow R \cup \{+\infty\}$ is defined by

$$I_S(x) := \begin{cases} 0, & \text{if } x \in S, \\ +\infty, & \text{otherwise.} \end{cases} \quad (1.3.9)$$

Clearly, I_S is convex if and only if S is convex.

Let $S \subset R^n$ be a nonempty subset. The support function of S is defined by

$$\sigma_S(s) := \sup\{s, x \mid x \in S\}. \quad (1.3.10)$$

This is a convex function.

It is easy to see that a norm on R^n is a convex function. If we define the distance function as

$$d_S(x) := \inf\{\|y - x\| \mid y \in S\},$$

where $S \subset R^n$ is a nonempty convex set and $\|\cdot\|$ is any norm on R^n , then d_S is a convex function.

A convex function can also be described by an epigraph. Now we first give the definition of the epigraph of f , and then show that f is convex if and only if its epigraph is a convex set.

Let $S \subset R^n$ be a nonempty set. A set $\{(x, f(x)) : x \in S\} \subset R^{n+1}$ describing the function f is said to be the graph of the function f . Related to the graph of f , there are the epigraph, which consists of points above the graph of f , and the hypograph, which consists of points below the graph of f .

Definition 1.3.7 Let $S \subset R^n$ be a nonempty set. Let $f : S \subset R^n \rightarrow R$. The epigraph of f , denoted by $\text{epi } f$, is a subset of R^{n+1} defined by

$$\text{epi } f = \{(x, \alpha) \mid f(x) \leq \alpha, x \in S, \alpha \in R\}. \quad (1.3.11)$$

The hypograph of f , denoted by $\text{hyp } f$, is a subset of R^{n+1} defined by

$$\text{hyp } f = \{(x, \alpha) \mid f(x) \geq \alpha, x \in S, \alpha \in R\}. \quad (1.3.12)$$

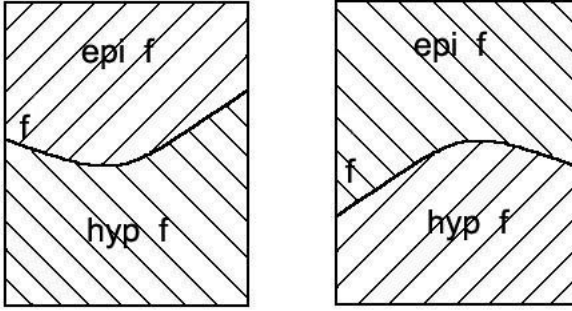


Figure 1.3.3 Epigraph and hypograph

The following theorem indicates the relation between convex function and convexity of $\text{epi } f$.

Theorem 1.3.8 Let $S \subset R^n$ be a nonempty convex set. Let $f : S \subset R^n \rightarrow R$. Then f is convex if and only if $\text{epi } f$ is a convex set.

Proof. Assume that f is convex. Let $x_1, x_2 \in S$ and $(x_1, \alpha_1), (x_2, \alpha_2)$ be in $\text{epi } f$. Then, it follows from Definition 1.3.6 and Definition 1.3.7 that

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2) \leq \lambda \alpha_1 + (1 - \lambda)\alpha_2$$

for any $\lambda \in (0, 1)$. Since S is a convex set, $\lambda x_1 + (1 - \lambda)x_2 \in S$. Hence $(\lambda x_1 + (1 - \lambda)x_2, \lambda \alpha_1 + (1 - \lambda)\alpha_2) \in \text{epi } f$, which means $\text{epi } f$ is convex.

Conversely, assume that $\text{epi } f$ is convex, and let $x_1, x_2 \in S$ and $(x_1, f(x_1)), (x_2, f(x_2)) \in \text{epi } f$. Then we have from the convexity of $\text{epi } f$ that

$$(\lambda x_1 + (1 - \lambda)x_2, \lambda f(x_1) + (1 - \lambda)f(x_2)) \in \text{epi } f, \text{ for } \lambda \in (0, 1).$$

This means

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2)$$

for each $\lambda \in (0, 1)$. Hence f is convex. $\square$

The epigraph $\text{epi} f$ of a function f is an important concept and it is used often in convex programming. Here we would like to mention its properties. The $\text{epi} f$ has a closed relation with the lower semi-continuity (l.s.c.) of f which is also very important, because, for a function to have a minimum, a very basic requirement is lower semi-continuity. We may recall that a function f is lower semi-continuous if, for each $x \in R^n$,

$$\liminf_{y \rightarrow x} f(y) \geq f(x). \quad (1.3.13)$$

The following theorem gives an equivalent property between $\text{epi} f$ and l.s.c.

Theorem 1.3.9 *For $f : R^n \rightarrow R \cup \{+\infty\}$, the following three statements are equivalent:*

1. f is lower semi-continuous on R^n ;
2. $\text{epi} f$ is a closed set in $R^n \times R$;
3. the level sets $L_r(f) = \{x \in R^n \mid f(x) \leq r, r \in R\}$ are closed for all $r \in R$.

Proof. (1) $\Rightarrow$ (2): Let $\{(y_k, r_k)\}$ be a sequence of $\text{epi} f$ converging to (x, r) for $k \rightarrow \infty$. Since $f(y_k) \leq r_k$ for all k , the (1.3.13) gives

$$r = \lim_{k \rightarrow \infty} r_k \geq \liminf_{y_k \rightarrow x} f(y_k) \geq f(x),$$

which indicates that $(x, r) \in \text{epi} f$.

(2) $\Rightarrow$ (3): Construct the level set $L_r(f)$ which is the intersection of two closed sets $\text{epi} f$ and $(R^n \times \{r\})$. Obviously the intersection is closed.

(3) $\Rightarrow$ (1): Suppose that f is not lower semi-continuous at some x , which means there exists a sequence $\{y_k\}$ converging to x such that $\{f(y_k)\}$ converges to $\rho < f(x) \leq +\infty$. Take $r \in (\rho, f(x))$. When k tends large enough, we have $f(y_k) \leq r < f(x)$ which means that $L_r(f)$ does not contain its limit x . Hence $L_r(f)$ is not closed. $\square$

Using Theorem 1.3.9, we can give a definition of closed function.

Definition 1.3.10 A function $f : R^n \rightarrow R \cup \{+\infty\}$ is said to be closed if it is lower semi-continuous everywhere, or if its epigraph is closed, or if its level sets are closed.

Obviously, the indicator function I_S is closed if and only if S is closed. Also, $\text{epi} I_S = S \times R^+$. The support function σ_S is closed too.

Next, we give some properties of convex functions.

Theorem 1.3.11 1. Let f be a convex function on a convex set $S \subset R^n$ and real number $\alpha \geq 0$, then αf is also a convex function on S .

2. Let f_1, f_2 be convex functions on a convex set S , then $f_1 + f_2$ is also a convex function on S .

3. Let $f_1, f_2, \dots, f_m$ be convex functions on a convex set S and real numbers $\alpha_1, \alpha_2, \dots, \alpha_m \geq 0$, then $\sum_{i=1}^m \alpha_i f_i$ is also a convex function on S .

Proof. We only prove the second statement. The others are similar.

Let $x_1, x_2 \in S$ and $0 < \alpha < 1$, then

$$\begin{aligned} & f_1(\alpha x_1 + (1 - \alpha)x_2) + f_2(\alpha x_1 + (1 - \alpha)x_2) \\ & \leq \alpha[f_1(x_1) + f_2(x_1)] + (1 - \alpha)[f_1(x_2) + f_2(x_2)]. \quad \square \end{aligned}$$

Continuity is an important property of a convex function. However, it is not sure that a convex function whose domain is not open is continuous. The following theorem shows that a convex function is continuous on an open convex set or the interior of its domain.

Theorem 1.3.12 Let $S \subset D$ be an open convex set. Let $f : D \subset R^n \rightarrow R$ be convex. Then f is continuous on S .

Proof. Let x_0 be an arbitrary point in S . Since S is an open convex set, we can find $n + 1$ points $x_1, \dots, x_{n+1} \in S$ such that the interior of the convex hull

$$C = \{x \mid x = \sum_{i=1}^{n+1} \alpha_i x_i, \alpha_i \geq 0, \sum_{i=1}^{n+1} \alpha_i = 1\}$$

is not empty and $x_0 \in \text{int} C$.

Now let $\alpha = \max_{1 \leq i \leq n+1} f(x_i)$, then

$$f(x) = f\left(\sum_{i=1}^{n+1} \alpha_i x_i\right) \leq \sum_{i=1}^{n+1} \alpha_i f(x_i) \leq \alpha, \quad \forall x \in C, \quad (1.3.14)$$

so that f is bounded over C . Also, since $x_0 \in \text{int } C$, there is a $\delta > 0$ such that $B(x_0, \delta) \subset C$, where $B(x_0, \delta) = \{x \mid \|x - x_0\| \leq \delta\}$. Hence for arbitrary $h \in B(0, \delta)$ and $\lambda \in [0, 1]$, we have

$$x_0 = \frac{1}{1+\lambda}(x_0 + \lambda h) + \frac{\lambda}{1+\lambda}(x_0 - h). \quad (1.3.15)$$

Since f is convex on C , then

$$f(x_0) \leq \frac{1}{1+\lambda}f(x_0 + \lambda h) + \frac{\lambda}{1+\lambda}f(x_0 - h). \quad (1.3.16)$$

By (1.3.16) and (1.3.14), we have

$$f(x_0 + \lambda h) - f(x_0) \geq \lambda(f(x_0) - f(x_0 - h)) \geq -\lambda(\alpha - f(x_0)). \quad (1.3.17)$$

On the other hand,

$$f(x_0 + \lambda h) = f(\lambda(x_0 + h) + (1 - \lambda)x_0) \leq \lambda f(x_0 + h) + (1 - \lambda)f(x_0),$$

which is

$$f(x_0 + \lambda h) - f(x_0) \leq \lambda(f(x_0 + h) - f(x_0)) \leq \lambda(\alpha - f(x_0)). \quad (1.3.18)$$

Therefore, (1.3.17) and (1.3.18) give

$$|f(x_0 + \lambda h) - f(x_0)| \leq \lambda|f(x_0) - \alpha|. \quad (1.3.19)$$

Now, for given $\epsilon > 0$, choose $\delta' \leq \delta$ so that $\delta'|f(x_0) - \alpha| \leq \epsilon\delta$. Set $d = \lambda h$ with $\|h\| = \delta$, then $d \in B(0, \delta)$ and

$$|f(x_0 + d) - f(x_0)| \leq \epsilon. \quad \square$$

If a convex function is differentiable, we can describe the characterization of differential convex functions. The following theorem gives the first order characterization of differential convex functions.

Theorem 1.3.13 *Let $S \subset R^n$ be a nonempty open convex set and let $f : S \subset R^n \rightarrow R$ be a differentiable function. Then f is convex if and only if*

$$f(y) \geq f(x) + \nabla f(x)^T(y - x), \quad \forall x, y \in S. \quad (1.3.20)$$

Similarly, f is strictly convex on S if and only if

$$f(y) > f(x) + \nabla f(x)^T(y - x), \quad \forall x, y \in S, y \neq x. \quad (1.3.21)$$

Furthermore, f is strongly (or uniformly) convex if and only if

$$f(y) \geq f(x) + \nabla f(x)^T(y - x) + \frac{1}{2}c\|y - x\|^2, \quad \forall x, y \in S, \quad (1.3.22)$$

where $c > 0$ is a constant.

Proof. Necessity: Let $f(x)$ be a convex function, then for all α with $0 < \alpha < 1$,

$$f(\alpha y + (1 - \alpha)x) \leq \alpha f(y) + (1 - \alpha)f(x).$$

Hence,

$$\frac{f(x + \alpha(y - x)) - f(x)}{\alpha} \leq f(y) - f(x).$$

Setting $\alpha \rightarrow 0$ yields

$$\nabla f(x)^T(y - x) \leq f(y) - f(x).$$

Sufficiency: Assume that (1.3.20) holds. Pick any $x_1, x_2 \in S$ and set $x = \alpha x_1 + (1 - \alpha)x_2, 0 < \alpha < 1$. Then

$$\begin{aligned} f(x_1) &\geq f(x) + \nabla f(x)^T(x_1 - x), \\ f(x_2) &\geq f(x) + \nabla f(x)^T(x_2 - x). \end{aligned}$$

Hence

$$\begin{aligned} \alpha f(x_1) + (1 - \alpha)f(x_2) &\geq f(x) + \nabla f(x)^T(\alpha x_1 + (1 - \alpha)x_2 - x) \\ &= f(\alpha x_1 + (1 - \alpha)x_2), \end{aligned}$$

which indicates that $f(x)$ is a convex function.

Similarly, we can prove (1.3.21) and (1.3.22) by use of (1.3.20). For example, from the definition of the strictly convex, we have

$$f(x + \alpha(y - x)) - f(x) < \alpha(f(y) - f(x)).$$

Then, using (1.3.20) and the above inequality, we have

$$\langle \nabla f(x), \alpha(y - x) \rangle \leq f(x + \alpha(y - x)) - f(x) < \alpha(f(y) - f(x))$$

which is the required (1.3.21).

To obtain (1.3.22), it is enough to apply (1.3.20) to the function $f - \frac{1}{2}c\|\cdot\|^2$.

□

Definition 1.3.6 of convex function indicates that the function value is below the chord, which means that the linear interpolation of the function values at two points is larger than the function value at the interpolation point. This theorem represents that the linear approximation based on a local derivative is a lower estimate, i.e., the convex function always lies above its tangent at any point. Such a tangent is called a supporting hyperplane of the convex function.

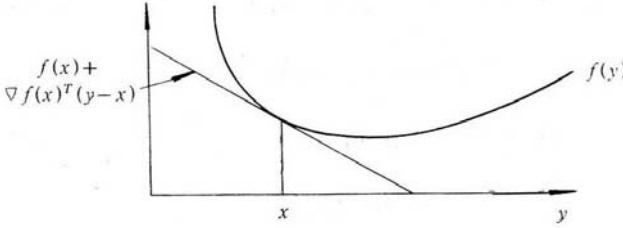


Figure 1.3.4 The first order characteristic of a convex function

Below, we consider the second order characteristic of a twice continuously differentiable convex function.

Theorem 1.3.14 *Let $S \subset \mathbb{R}^n$ be a nonempty open convex set, and let $f : S \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be twice continuously differentiable. Then*

1. *f is convex if and only if its Hessian matrix is positive semidefinite at each point in S .*
2. *f is strictly convex if its Hessian matrix is positive definite at each point in S .*
3. *f is uniformly convex if and only if its Hessian matrix is uniformly positive definite at each point in S , i.e., there exists a constant $m > 0$ such that*

$$m\|u\|^2 \leq u^T \nabla^2 f(x)u, \quad \forall x \in S, u \in \mathbb{R}^n.$$

Proof. We only prove the first case. The other two cases are analogous.

Sufficiency. Suppose that the Hessian matrix $\nabla^2 f(x)$ is positive semidefinite at each point $x \in S$. Consider $x, \bar{x} \in S$. By the mean-value theorem, we have

$$f(x) = f(\bar{x}) + \nabla f(\bar{x})^T(x - \bar{x}) + \frac{1}{2}(x - \bar{x})^T \nabla^2 f(\hat{x})(x - \bar{x}),$$

where $\hat{x} = \bar{x} + \theta(x - \bar{x})$, $\theta \in (0, 1)$. Noting that $\hat{x} \in S$, it follows from the assumption that

$$f(x) \geq f(\bar{x}) + \nabla f(\bar{x})^T(x - \bar{x}).$$

Hence f is a convex function by Theorem 1.3.13.

Necessity. Suppose that f is a convex function and let $\bar{x} \in S$. We need to prove $p^T \nabla^2 f(\bar{x})p \geq 0$, $\forall p \in R^n$. Since S is open, then there exists $\delta > 0$ such that when $|\lambda| < \delta$, $\bar{x} + \lambda p \in S$. By Theorem 1.3.13,

$$f(\bar{x} + \lambda p) \geq f(\bar{x}) + \lambda \nabla f(\bar{x})^T p. \quad (1.3.23)$$

Also since $f(x)$ is twice differentiable at $\bar{x}$, then

$$f(\bar{x} + \lambda p) = f(\bar{x}) + \lambda \nabla f(\bar{x})^T p + \frac{\lambda^2}{2} p^T \nabla^2 f(\bar{x})p + o(\|\lambda p\|^2). \quad (1.3.24)$$

Substituting (1.3.24) into (1.3.23) yields

$$\frac{1}{2} \lambda^2 p^T \nabla^2 f(\bar{x})p + o(\|\lambda p\|^2) \geq 0.$$

Dividing by λ^2 and letting $\lambda \rightarrow 0$, it follows that

$$p^T \nabla^2 f(\bar{x})p \geq 0. \square$$

Next, we would like to characterize a convex function with monotonicity which is very useful.

We first introduce a definition of monotone mapping.

Definition 1.3.15 A mapping $F : D \subset R^n \rightarrow R^n$ is monotone on $D_0 \subset D$ if

$$\langle F(x) - F(y), x - y \rangle \geq 0, \quad \forall x, y \in D_0; \quad (1.3.25)$$

F is strictly monotone on D_0 if

$$\langle F(x) - F(y), x - y \rangle > 0, \quad \forall x, y \in D_0, x \neq y; \quad (1.3.26)$$

F is uniformly (or strongly) monotone if there is a constant $c > 0$ so that

$$\langle F(x) - F(y), x - y \rangle \geq c\|x - y\|^2, \quad \forall x, y \in D_0. \quad (1.3.27)$$

If we let $F = \nabla f$ in the above definition, we can get the following theorem which says that, for convex function f , its gradient ∇f is a monotone mapping.

Theorem 1.3.16 *Suppose that $f : D \subset R^n \rightarrow R$ is differentiable on an open set D , and that $S \subset D$ is a convex subset. Then f is convex on S if and only if its gradient ∇f is monotone, i.e.,*

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq 0, \quad \forall x, y \in S; \quad (1.3.28)$$

and f is strictly convex on S if and only if its gradient ∇f is strictly monotone, i.e.,

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle > 0, \quad \forall x, y \in S, x \neq y; \quad (1.3.29)$$

finally, f is uniformly (or strongly) convex on S if and only if its gradient ∇f is uniformly monotone, i.e.,

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq c\|x - y\|^2, \quad (1.3.30)$$

where $c > 0$ is the constant of (1.3.8).

Proof. Let f be uniformly convex on S , then, by Theorem 1.3.13, for any $x, y \in S$, we have

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{1}{2}c\|y - x\|^2, \quad (1.3.31)$$

$$f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle + \frac{1}{2}c\|x - y\|^2, \quad (1.3.32)$$

and addition of these two inequalities shows that (1.3.30) holds.

Similarly, if f is convex, (1.3.31) and (1.3.32) hold with $c = 0$, and hence (1.3.28) holds. Moreover, if f is strictly convex, then (1.3.31) and (1.3.32) hold with $c = 0$ but with strict inequality for $x \neq y$. Hence the addition establishes (1.3.29).

Conversely, suppose that ∇f is monotone. For any fixed $x, y \in S$, the mean-value theorem (1.2.97) gives

$$f(y) - f(x) = \langle \nabla f(\xi), y - x \rangle, \quad (1.3.33)$$

where $\xi = x + t(y - x)$, $t \in (0, 1)$. Then, it follows from (1.3.28) that

$$\langle \nabla f(\xi) - \nabla f(x), y - x \rangle = \frac{1}{t} [\nabla f(\xi) - \nabla f(x)]^T (\xi - x) \geq 0, \quad (1.3.34)$$

which, together with (1.3.33), gives

$$\begin{aligned} f(y) - f(x) &= \langle \nabla f(\xi) - \nabla f(x), y - x \rangle + \langle \nabla f(x), y - x \rangle \\ &\geq \langle \nabla f(x), y - x \rangle. \end{aligned} \quad (1.3.35)$$

The above inequality shows, by Theorem 1.3.13, that f is convex.

Similarly, if (1.3.29) holds, the same will be true in (1.3.35) with strict inequality and $x \neq y$, and thus f is strictly convex.

Finally, for uniform convexity, suppose (1.3.30) holds. Let $\phi(t) = f(x + t(y - x)) = f(u)$, where $u = x + t(y - x)$, $t \in (0, 1)$. Noting that $\phi'(t) = \langle \nabla f(u), y - x \rangle$ and $\phi'(0) = \langle \nabla f(x), y - x \rangle$, then (1.3.30) means

$$\begin{aligned} \phi'(t) - \phi'(0) &= \langle \nabla f(u) - \nabla f(x), y - x \rangle = \frac{1}{t} \langle \nabla f(u) - \nabla f(x), u - x \rangle \\ &\geq \frac{1}{t} c \|u - x\|^2 = tc \|y - x\|^2. \end{aligned}$$

Hence,

$$\phi(1) - \phi(0) - \phi'(0) = \int_0^1 [\phi'(t) - \phi'(0)] dt \geq \frac{1}{2} c \|y - x\|^2,$$

which, by the definition of ϕ , shows

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{1}{2} c \|y - x\|^2.$$

Therefore, we complete the proof. $\square$

Combining Theorem 1.3.14 and 1.3.16, we immediately obtain the following theorem.

Theorem 1.3.17 *Let $S \subset R^n$ be a nonempty open convex set and f be a twice continuously differentiable function on S . Then*

1. ∇f is monotone on S if and only if $\nabla^2 f(x)$ is positive semidefinite for all $x \in S$.

2. If $\nabla^2 f(x)$ is positive definite for all $x \in S$, then ∇f is strictly monotone on S .
3. ∇f is uniformly (or strongly) monotone on S if and only if $\nabla^2 f(x)$ is uniformly positive definite, i.e., there exists a number $c > 0$ so that

$$d^T \nabla^2 f(x) d \geq c \|d\|^2, \quad \forall x \in S, d \in R^n.$$

In the following, we are concerned with the level set which is closely related to a convex function and important to the minimization algorithm. The following theorem shows that the level set L_α corresponding to a convex function is convex.

Theorem 1.3.18 *Let $S \subset R^n$ be a nonempty convex set, f a convex function defined on S , α a real number. Then the level set $L_\alpha = \{x \mid x \in S, f(x) \leq \alpha\}$ is a convex set.*

Proof. Let $x_1, x_2 \in L_\alpha$, then $x_1, x_2 \in S, f(x_1) \leq \alpha, f(x_2) \leq \alpha$. Let $\lambda \in (0, 1)$ and $x = \lambda x_1 + (1 - \lambda)x_2$. Then from the convexity of S , we have $x \in S$. Also since f is convex,

$$f(x) \leq \lambda f(x_1) + (1 - \lambda)f(x_2) \leq \lambda\alpha + (1 - \lambda)\alpha = \alpha.$$

Hence $x \in L_\alpha$, which implies that L_α is a convex set. $\square$

From Theorem 1.3.18 and Theorem 1.3.9, we know immediately that if f is a continuously convex function, then the level set L_α is a closed convex set. Furthermore, we also have

Theorem 1.3.19 *Let $f(x)$ be twice continuously differentiable on $S \subset R^n$, where S is a nonempty convex set. Suppose that there exists a number $m > 0$ such that*

$$u^T \nabla^2 f(x) u \geq m \|u\|^2, \quad \forall x \in L(x_0), u \in R^n. \quad (1.3.36)$$

Then the level set $L(x_0) = \{x \in S \mid f(x) \leq f(x_0)\}$ is a bounded closed convex set.

Proof. By using Theorem 1.3.14, (1.3.36) implies that f is convex on $L(x_0)$, and then it follows from Theorem 1.3.18 that $L(x_0)$ is convex. Note that $f(x)$ is continuous, then $L(x_0)$ is a closed convex set for all $x_0 \in R^n$.

Now we prove the boundedness of $L(x_0)$. Using (1.3.36) and the fact that $L(x_0)$ is convex, we have for any $x, y \in L(x_0)$,

$$m\|y - x\|^2 \leq (y - x)^T \nabla^2 f(x + \alpha(y - x))(y - x).$$

Also by twice differentiability and the above inequality, we have

$$\begin{aligned} f(y) &= f(x) + \nabla f(x)^T(y - x) \\ &\quad + \int_0^1 \int_0^t (y - x)^T \nabla^2 f(x + \alpha(y - x))(y - x) d\alpha dt \\ &\geq f(x) + \nabla f(x)^T(y - x) + \frac{1}{2}m\|y - x\|^2, \end{aligned}$$

where m is independent of x and y . Therefore for arbitrary $y \in L(x_0)$ and $y \neq x_0$,

$$\begin{aligned} f(y) - f(x_0) &\geq \nabla f(x_0)^T(y - x_0) + \frac{1}{2}m\|y - x_0\|^2 \\ &\geq -\|\nabla f(x_0)\|\|y - x_0\| + \frac{1}{2}m\|y - x_0\|^2. \end{aligned}$$

Noting that $f(y) \leq f(x_0)$, the above inequality implies

$$\|y - x_0\| \leq \frac{2}{m}\|\nabla f(x_0)\|,$$

which shows that the level set $L(x_0)$ is bounded. $\square$

To conclude this subsection, we give a proof of Minkowski inequality which is an application of convexity of function.

Minkowski inequality:

$$\|x + y\|_p \leq \|x\|_p + \|y\|_p, \quad (1.3.37)$$

i.e.,

$$\left(\sum_{i=1}^n |x_i + y_i|^p \right)^{1/p} \leq \left(\sum_{i=1}^n |x_i|^p \right)^{1/p} + \left(\sum_{i=1}^n |y_i|^p \right)^{1/p}, \quad (1.3.38)$$

where $p \geq 1$.

Proof. If x or y is the zero vector, the result is obvious. Now suppose that $x \neq 0$ and $y \neq 0$.

If $p = 1$, since $|x_i + y_i| \leq |x_i| + |y_i|, i = 1, \dots, n$, then summing over i gives the result.

Now let $p > 1$ and consider the function

$$\phi(t) = t^p, \quad t > 0.$$

Since

$$\phi''(t) = p(p-1)t^{p-2} > 0,$$

then $\phi(t)$ is strictly convex. Note that

$$\frac{\|x\|_p}{\|x\|_p + \|y\|_p} + \frac{\|y\|_p}{\|x\|_p + \|y\|_p} = 1,$$

it follows from the definition of convex function that

$$\begin{aligned} & \left(\frac{\|x\|_p}{\|x\|_p + \|y\|_p} \frac{|x_i|}{\|x\|_p} + \frac{\|y\|_p}{\|x\|_p + \|y\|_p} \frac{|y_i|}{\|y\|_p} \right)^p \\ & \leq \frac{\|x\|_p}{\|x\|_p + \|y\|_p} \left(\frac{|x_i|}{\|x\|_p} \right)^p + \frac{\|y\|_p}{\|x\|_p + \|y\|_p} \left(\frac{|y_i|}{\|y\|_p} \right)^p. \end{aligned} \quad (1.3.39)$$

Hence, using (1.3.39), we get

$$\begin{aligned} & \sum_{i=1}^n \left(\frac{|x_i + y_i|}{\|x\|_p + \|y\|_p} \right)^p \\ & \leq \sum_{i=1}^n \left(\frac{|x_i| + |y_i|}{\|x\|_p + \|y\|_p} \right)^p \\ & = \sum_{i=1}^n \left(\frac{\|x\|_p}{\|x\|_p + \|y\|_p} \frac{|x_i|}{\|x\|_p} + \frac{\|y\|_p}{\|x\|_p + \|y\|_p} \frac{|y_i|}{\|y\|_p} \right)^p \\ & \leq \sum_{i=1}^n \left(\frac{\|x\|_p}{\|x\|_p + \|y\|_p} \left(\frac{|x_i|}{\|x\|_p} \right)^p + \frac{\|y\|_p}{\|x\|_p + \|y\|_p} \left(\frac{|y_i|}{\|y\|_p} \right)^p \right) \\ & \leq \frac{\|x\|_p}{\|x\|_p + \|y\|_p} \sum_{i=1}^n \left(\frac{|x_i|}{\|x\|_p} \right)^p + \frac{\|y\|_p}{\|x\|_p + \|y\|_p} \sum_{i=1}^n \left(\frac{|y_i|}{\|y\|_p} \right)^p \\ & = \frac{\|x\|_p}{\|x\|_p + \|y\|_p} \frac{\|x\|_p^p}{\|x\|_p^p} + \frac{\|y\|_p}{\|x\|_p + \|y\|_p} \frac{\|y\|_p^p}{\|y\|_p^p} \\ & = 1, \end{aligned}$$

which implies that

$$\sum_{i=1}^n |x_i + y_i|^p \leq (\|x\|_p + \|y\|_p)^p.$$

Taking the p -th root gives our result. $\square$

1.3.3 Separation and Support of Convex Sets

The separation and support of convex sets are important tools for research of optimality conditions. We first discuss the projection theorem which characterizes the projection and describes the sufficient and necessary condition for the distance between a closed convex set and a point not in the set to be minimal.

Theorem 1.3.20 (*Projection Theorem*)

Let $S \subset R^n$ be a nonempty closed convex set and $y \notin S$, then there exists a unique point $\bar{x} \in S$ with minimal distance from y , i.e.,

$$\|y - \bar{x}\| = \inf_{x \in S} \|y - x\|. \quad (1.3.40)$$

Furthermore, $\bar{x}$ is the minimal point of (1.3.40) if and only if

$$\langle y - \bar{x}, x - \bar{x} \rangle \leq 0, \quad \forall x \in S, \quad (1.3.41)$$

or say that $\bar{x}$ is the projection $P_S(y)$ of y on S if and only if (1.3.41) holds.

Proof. Let

$$\inf\{\|y - x\| \mid x \in S\} = \gamma > 0. \quad (1.3.42)$$

There is a sequence $\{x_k\} \subset S$ so that $\|y - x_k\| \rightarrow \gamma$. In the following, we prove $\{x_k\}$ is a Cauchy sequence and hence there exists a limit $\bar{x} \in S$.

By the parallelogram law, we have

$$\begin{aligned} \|x_k - x_m\|^2 &= 2\|x_k - y\|^2 + 2\|x_m - y\|^2 - \|x_k + x_m - 2y\|^2 \\ &= 2\|x_k - y\|^2 + 2\|x_m - y\|^2 - 4\left\|\frac{x_k + x_m}{2} - y\right\|^2 \end{aligned} \quad (1.3.43)$$

Note that $(x_k + x_m)/2 \in S$, we have, from the definition of γ ,

$$\left\|\frac{x_k + x_m}{2} - y\right\|^2 \geq \gamma^2.$$

Therefore,

$$\|x_k - x_m\|^2 \leq 2\|x_k - y\|^2 + 2\|x_m - y\|^2 - 4\gamma^2.$$

Taking k and m sufficiently large yields

$$\|x_k - x_m\| \rightarrow 0$$

which indicates that $\{x_k\}$ is a Cauchy sequence with limit $\bar{x}$. Since S is closed, then $\bar{x} \in S$. This shows there exists $\bar{x}$ such that $\|y - \bar{x}\| = \gamma$.

Next, we prove the uniqueness. Suppose that $\bar{x}, \bar{x}' \in S$ and satisfy

$$\|y - \bar{x}\| = \|y - \bar{x}'\| = \gamma. \quad (1.3.44)$$

Since S is convex, $(\bar{x} + \bar{x}')/2 \in S$. Then

$$\left\| y - \frac{\bar{x} + \bar{x}'}{2} \right\| \leq \frac{1}{2}\|y - \bar{x}\| + \frac{1}{2}\|y - \bar{x}'\| = \gamma. \quad (1.3.45)$$

If the strict inequality holds, we get a contradiction to (1.3.42). Then the equality holds in (1.3.45) and we have

$$y - \bar{x} = \lambda(y - \bar{x}'), \text{ for some } \lambda.$$

So, it follows from (1.3.44) that $|\lambda| = 1$. If $\lambda = -1$, we have $y = (\bar{x} + \bar{x}')/2 \in S$ which contradicts $y \notin S$. Therefore, $\lambda = 1$, that means $\bar{x} = \bar{x}'$.

Finally, we prove that the distance between $\bar{x} \in S$ and $y \notin S$ is minimal if and only if (1.3.41) holds.

Take x arbitrary in S and suppose (1.3.41) holds. Since

$$\begin{aligned} \|y - x\|^2 &= \|y - \bar{x} + \bar{x} - x\|^2 \\ &= \|y - \bar{x}\|^2 + \|\bar{x} - x\|^2 + 2(\bar{x} - x)^T(y - \bar{x}), \end{aligned}$$

then $\|y - x\|^2 \geq \|y - \bar{x}\|^2$ which is the desired sufficiency.

Conversely, let $\|y - x\|^2 \geq \|y - \bar{x}\|^2$, $\forall x \in S$. Since $\bar{x} + \lambda(x - \bar{x}) \in S$ with $\lambda \in (0, 1)$, then we have

$$\|y - \bar{x} - \lambda(x - \bar{x})\|^2 \geq \|y - \bar{x}\|^2.$$

Developing the square gives

$$\|y - \bar{x} - \lambda(x - \bar{x})\|^2 = \|y - \bar{x}\|^2 + \lambda^2\|x - \bar{x}\|^2 + 2\lambda(x - \bar{x})^T(\bar{x} - y).$$

Then we get

$$\lambda^2\|x - \bar{x}\|^2 + 2\lambda(x - \bar{x})^T(\bar{x} - y) \geq 0.$$

Dividing by λ and letting $\lambda \downarrow 0$, we obtain the result. $\square$

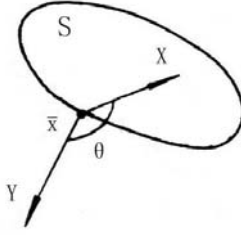


Figure 1.3.5 The angle-characterization of a projection

In fact, if we note that $\bar{x} = P_S(y)$ is the solution of

$$\min_{x \in S} \phi(x) \triangleq \frac{1}{2} \|x - y\|^2,$$

then, for the above minimization problem, one concludes from the optimality condition that

$$\langle \phi'(\bar{x}), x - \bar{x} \rangle \geq 0, \quad \forall x \in S.$$

Since $\phi'(x) = x - y$, we have

$$\langle \bar{x} - y, x - \bar{x} \rangle \geq 0, \quad \forall x \in S$$

which is just (1.3.41).

Remark: If S is an affine manifold (for example, a subspace), then $\bar{x} - x \in S$ whenever $x - \bar{x} \in S$. Therefore, (1.3.41) implies

$$\langle y - \bar{x}, x - \bar{x} \rangle = 0, \quad \forall x \in S, \quad (1.3.46)$$

which is $(y - \bar{x}) \perp S$.

Now we can present the most fundamental separation theorem which separates a closed convex set and a point not in the set. This theorem is based on the above projection theorem.

Theorem 1.3.21 *Let $S \subset R^n$ be a nonempty closed convex set and $y \notin S$. Then there exist a nonzero vector p and a real number α such that*

$$p^T y > \alpha \text{ and } p^T x \leq \alpha, \quad \forall x \in S, \quad (1.3.47)$$

i.e.,

$$p^T y > \sup\{p^T x, \forall x \in S\} \quad (1.3.48)$$

which says there exists a hyperplane $H = \{x \mid p^T x = \alpha\}$ that strictly separates y and S .

Proof. Since S is a nonempty closed convex set and $y \notin S$, then, by Theorem 1.3.20, there exists a unique point $\bar{x} \in S$ such that

$$(x - \bar{x})^T(y - \bar{x}) \leq 0, \quad \forall x \in S.$$

Set $p = y - \bar{x} \neq 0$, then

$$\begin{aligned} 0 &\geq (y - \bar{x})^T(y - \bar{x} + x - y) \\ &= p^T x - p^T y + \|p\|^2. \end{aligned}$$

Hence

$$p^T y \geq p^T x + \|p\|^2, \quad \forall x \in S.$$

Set $\alpha = \sup\{p^T x \mid x \in S\}$, and we get our result. $\square$

As a consequence of Theorem 1.3.21, we immediately obtain Farkas' Lemma which has been used extensively in the derivation of optimality conditions.

Theorem 1.3.22 (*Farkas' Lemma*) *Let $A \in R^{m \times n}$ and $c \in R^n$. Then exactly one of the following two systems has a solution:*

$$\text{System 1} \quad Ax \leq 0, \quad c^T x > 0, \quad (1.3.49)$$

$$\text{System 2} \quad A^T y = c, \quad y \geq 0. \quad (1.3.50)$$

Proof. Suppose that there is a solution for (1.3.50), that is, there exists $y \geq 0$ such that $A^T y = c$. Let x satisfy $Ax \leq 0$, it follows from $y \geq 0$ that

$$c^T x = y^T Ax \leq 0,$$

which shows that (1.3.49) has no solution.

Now suppose (1.3.50) has no solution. Let

$$S = \{x \mid x = A^T y, y \geq 0\},$$

which is a polyhedral set, and hence it is a nonempty closed convex set and $c \notin S$. By Theorem 1.3.21, there exist $p \in R^n$ and $\alpha \in R$ such that

$$p^T c > \alpha \text{ and } p^T x \leq \alpha, \quad \forall x \in S.$$

Since $0 \in S$, $\alpha \geq p^T 0 = 0$. Then $p^T c > 0$. Also note that

$$\alpha \geq p^T x = p^T A^T y = y^T A p, \quad \forall y \geq 0$$

and that y could be made arbitrarily large, thus it follows that $Ap \leq 0$. So, there is a vector $p \in R^n$ which is a solution of (1.3.49), and the proof is complete. $\square$

In order to discuss the separation between two convex sets, we need the following definition and theorem of a supporting hyperplane.

Definition 1.3.23 *Let $S \subset R^n$ be a nonempty set, $p \in R^n$, and $\bar{x} \in \partial S$, where ∂S denotes the boundary of S . If either*

$$S \subset H^+ = \{x \in S \mid p^T(x - \bar{x}) \geq 0\} \quad (1.3.51)$$

or

$$S \subset H^- = \{x \in S \mid p^T(x - \bar{x}) \leq 0\}, \quad (1.3.52)$$

then the hyperplane $H = \{x \in S \mid p^T(x - \bar{x}) = 0\}$ is called a supporting hyperplane of S at $\bar{x}$. If, in addition, $S \not\subset H$, then H is called a proper supporting hyperplane of S at $\bar{x}$.

The following theorem shows that a convex set has a supporting hyperplane at each boundary point (see Figure 1.3.6).

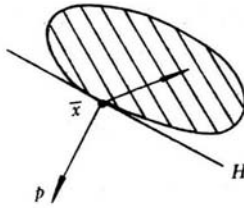


Figure 1.3.6 Supporting hyperplane

Theorem 1.3.24 *Let $S \subset R^n$ be a nonempty convex set and $\bar{x} \in \partial S$. Then, there exists a hyperplane supporting S at $\bar{x}$; that is, there exists a nonzero vector p such that*

$$p^T(x - \bar{x}) \leq 0, \quad \forall x \in \bar{S}, \quad (1.3.53)$$

where $\bar{S}$ denotes the closure of S .

Proof. Since $\bar{x} \in \partial S$, there exists a sequence $\{y_k\} \not\subset \bar{S}$ so that $y_k \rightarrow \bar{x}, (k \rightarrow \infty)$. By Theorem 1.3.21, corresponding to each y_k , there exists $p_k \in R^n$ with $\|p_k\| = 1$, such that

$$p_k^T y_k > p_k^T x, \forall x \in \bar{S}. \quad (1.3.54)$$

Since $\{p_k\}$ is bounded, there is a convergent subsequence $\{p_k\}_{\mathcal{K}}$ with limit p and $\|p\| = 1$. For this subsequence, (1.3.54) holds, that is

$$p_{k_j}^T y_{k_j} > p_{k_j}^T x, \forall x \in \bar{S}.$$

Fix $x \in \bar{S}$ and take the limit as $k \in \mathcal{K}$ and $k \rightarrow \infty$, we have $p^T \bar{x} \geq p^T x, \forall x \in \bar{S}$, which is our desired result. $\square$

By use of Theorem 1.3.21 and Theorem 1.3.24, the following corollary is obvious.

Corollary 1.3.25 *Let $S \subset R^n$ be a nonempty convex set and $\bar{x} \notin S$. Then there exists a nonzero vector p such that*

$$p^T(x - \bar{x}) \leq 0, \forall x \in \bar{S}. \quad (1.3.55)$$

Proof. Let $\bar{x} \notin S$; there are two cases. If $\bar{x} \notin \bar{S}$, the conclusion is immediate from Theorem 1.3.21. If $\bar{x} \in \partial S$, the corollary reduces to Theorem 1.3.24. $\square$

Now, we are going to discuss the separation theorems of two convex sets which include separation theorem, strict separation theorem and strong separation theorem.

Definition 1.3.26 *Let $S_1, S_2 \subset R^n$ be nonempty convex sets. If*

$$p^T x \geq \alpha, \forall x \in S_1 \text{ and } p^T x \leq \alpha, \forall x \in S_2, \quad (1.3.56)$$

then the hyperplane $H = \{x \mid p^T x = \alpha\}$ is said to separate S_1 and S_2 . If, in addition, $S_1 \cup S_2 \not\subset H$, then H is said to properly separate S_1 and S_2 . If

$$p^T x > \alpha, \forall x \in S_1 \text{ and } p^T x < \alpha, \forall x \in S_2, \quad (1.3.57)$$

then H is said to strictly separate S_1 and S_2 . If

$$p^T x \geq \alpha + \epsilon, \forall x \in S_1 \text{ and } p^T x \leq \alpha, \forall x \in S_2, \quad (1.3.58)$$

then H is said to strongly separate S_1 and S_2 , where $\epsilon > 0$.

Theorem 1.3.27 (*Separation Theorem*)

Let $S_1, S_2 \subset R^n$ be nonempty convex sets. If $S_1 \cap S_2 = \phi$, then there exists a hyperplane separating S_1 and S_2 , that is, there exists a nonzero vector $p \in R^n$ such that

$$p^T x_1 \leq p^T x_2, \forall x_1 \in \bar{S}_1, x_2 \in \bar{S}_2. \quad (1.3.59)$$

Proof. Let

$$S = S_1 - S_2 = \{x_1 - x_2 \mid x_1 \in S_1, x_2 \in S_2\}.$$

Note that S is a nonempty convex set and that $0 \notin S$ (otherwise, if $0 \in S$, then we have $x_1 - x_2 = 0$ and $x_1 = x_2 \in S_1 \cap S_2$ which implies $S_1 \cap S_2 \neq \phi$). Hence, by Corollary 1.3.25, there exists a nonzero vector p such that

$$p^T x \leq p^T 0 = 0, \forall x \in \bar{S},$$

which implies that

$$p^T x_1 \leq p^T x_2, \forall x_1 \in \bar{S}_1, x_2 \in \bar{S}_2.$$

Then we complete the proof. $\square$

Note that (1.3.59) also can be written as

$$\sup\{p^T x \mid x \in S_1\} \leq \inf\{p^T x \mid x \in S_2\}. \quad (1.3.60)$$

Theorem 1.3.28 (*Strong Separation Theorem*)

Let S_1 and S_2 be two closed convex sets on R^n , and suppose that S_2 is bounded. If $S_1 \cap S_2 = \phi$, then there exists a hyperplane that strongly separates S_1 and S_2 , that is, there exist a nonzero vector p and $\epsilon > 0$ such that

$$\inf\{p^T x \mid x \in S_2\} \geq \sup\{p^T x \mid x \in S_1\} + \epsilon. \quad (1.3.61)$$

Proof. Let $S = S_1 - S_2$. Note that S is convex and $0 \notin S$. We first prove that S is closed. Let $\{x_k\} \subset S, x_k \rightarrow x$. By the definition of S , $x_k = y_k - z_k, y_k \in S_1, z_k \in S_2$. Since S_2 is compact, there exists a convergent subsequence $\{z_k\}_{\mathcal{K}}, z_k \rightarrow z, z \in S_2, k \in \mathcal{K}$. Since

$$y_k - z_k \rightarrow x, z_k \rightarrow z, \forall k \in \mathcal{K},$$

then $y_k \rightarrow y$. Also since S_1 is closed, $y \in S_1$. Therefore,

$$x = y - z, y \in S_1, z \in S_2,$$

which means $x \in S$ and S is closed.

Now we have that S is a closed convex set and $0 \notin S$. Then, by Theorem 1.3.21, there exist nonzero vector p and real number α , such that

$$p^T x \leq \alpha, \forall x \in S \text{ and } p^T 0 > \alpha.$$

Hence, $\alpha < 0$. Using the definition of S yields

$$p^T z \geq p^T y - \alpha, \forall y \in S_1, z \in S_2,$$

which, by setting $\epsilon = -\alpha > 0$, is

$$\inf\{p^T z \mid z \in S_2\} \geq \sup\{p^T y \mid y \in S_1\} + \epsilon. \quad \square$$

Similarly, we can obtain the following strict separation theorem.

Theorem 1.3.29 (*Strict Separation Theorem*)

Let S_1 and S_2 be two closed convex sets on R^n , and suppose that S_2 is bounded. If $S_1 \cap S_2 = \emptyset$, there exists a nonzero vector p such that

$$\inf\{p^T x \mid x \in S_2\} > \sup\{p^T x \mid x \in S_1\}. \quad (1.3.62)$$

Proof. The result (1.3.62) is immediate from (1.3.61). $\square$

1.4 Optimality Conditions for Unconstrained Optimization

In this section we consider the unconstrained optimization problem

$$\min f(x), x \in R^n \quad (1.4.1)$$

and present its optimality conditions which include first-order conditions and second-order conditions.

In general, we have two types of minimizers: local minimizer and global minimizer. In the following, we give their exact definitions.

Definition 1.4.1 A point x^* is called a local minimizer if there exists $\delta > 0$ such that $f(x^*) \leq f(x)$ for all $x \in R^n$ satisfying $\|x - x^*\| < \delta$.

A point x^* is called a strict local minimizer if there exists $\delta > 0$ such that $f(x^*) < f(x)$ for all $x \in R^n$ with $x \neq x^*$ and $\|x - x^*\| < \delta$.

Definition 1.4.2 A point x^* is called a global minimizer if $f(x^*) \leq f(x)$ for all $x \in R^n$. A point x^* is called a strict global minimizer if $f(x^*) < f(x)$ for all $x \in R^n$ with $x \neq x^*$.

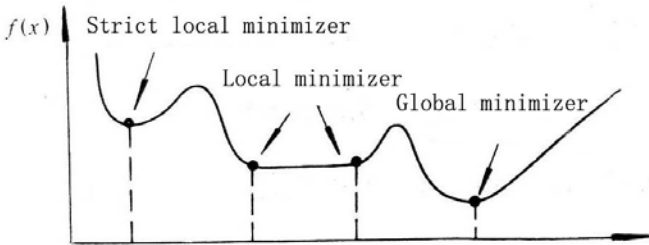


Figure 1.4.1 Types of minimal points

Note that, in practice, most algorithms are able to find only a local minimizer that is not a global minimizer. Normally, finding a global minimizer is a difficult task. In many practical applications, we are content with getting a local minimizer. In addition, many global optimization algorithms proceed by solving a sequence of local optimization algorithms. Hence, in this book, our focus is on the model, property, convergence and computation of local optimization algorithms. Usually, in the book, the minimizer refers to the local minimizer. However, if the problem is a convex programming problem, all local minimizers are also global minimizers.

The descent direction given in the following definition is an important concept.

Definition 1.4.3 Let $f : R^n \rightarrow R$ be differentiable at $x \in R^n$. If there exists a vector $d \in R^n$ such that

$$\langle \nabla f(x), d \rangle < 0, \quad (1.4.2)$$

then d is called a descent direction of f at x .

By means of Taylor's expansion,

$$f(x_k + td) = f(x_k) + t\nabla f(x_k)^T d + o(t),$$

then it is easy to see that

$$\exists \delta > 0 \text{ such that } f(x_k + td) < f(x_k), \forall t \in (0, \delta)$$

if and only if d is a descent direction of f at x_k .

Now we discuss the first-order optimality condition.

Theorem 1.4.4 (*First-Order Necessary Condition*)

Let $f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be continuously differentiable on an open set D . If $x^* \in D$ is a local minimizer of (1.4.1), then

$$\nabla f(x^*) = 0. \quad (1.4.3)$$

Proof. [proof I] Let x^* be a local minimizer. Consider the sequence

$$x_k = x^* - \alpha_k \nabla f(x^*), \quad \alpha_k > 0.$$

By Taylor's expansion, for k sufficiently large, we have

$$0 \leq f(x_k) - f(x^*) = -\alpha_k \nabla f(\eta_k)^T \nabla f(x^*),$$

where η_k is a convex combination of x_k and x^* . Dividing by α_k and taking the limit, it follows from $f \in C^1$ that

$$0 \leq -\|\nabla f(x^*)\|^2$$

which means $\nabla f(x^*) = 0$. $\square$

[proof II] (By contradiction). Suppose that $\nabla f(x^*) \neq 0$. Taking $d = -\nabla f(x^*)$ yields

$$d^T \nabla f(x^*) = -\|\nabla f(x^*)\|^2 < 0.$$

So, d is a descent direction and there exists $\delta > 0$ such that

$$f(x^* + \alpha d) < f(x^*), \quad \forall \alpha \in (0, \delta)$$

which contradicts the assumption that x^* is a local minimizer. $\square$

[proof III] Let x^* be a local minimizer, then there exists $\delta > 0$ so that $f(x) \geq f(x^*)$ for any x with $\|x - x^*\| < \delta$. By Taylor's expansion,

$$f(x) = f(x^*) + \nabla f(x^*)^T(x - x^*) + o(\|x - x^*\|) \geq f(x^*).$$

Dividing $\|x - x^*\|$ and letting $x \rightarrow x^*$ yield

$$\nabla f(x^*)^T \frac{(x - x^*)}{\|x - x^*\|} \geq 0.$$

Setting $s = (x - x^*)/\|x - x^*\|$, the above inequality is

$$\nabla f(x^*)^T s \geq 0, \quad \forall s \text{ with } \|s\| = 1.$$

Choosing $s = \pm e_i$, ($i = 1, \dots, n$), we obtain $\nabla f(x^*) = 0$. $\square$

Theorem 1.4.4 says that if x^* is a local minimizer, f has a zero slope at x^* . The following theorem indicates that if x^* is a local minimizer, f has nonnegative curvature at x^* besides zero slope.

Theorem 1.4.5 (*Second-Order Necessary Condition*)

Let $f : D \subset R^n \rightarrow R$ be twice continuously differentiable on an open set D . If x^* is a local minimizer of (1.4.1), then $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is positive semidefinite.

Proof. [proof I] We have known from Theorem 1.4.4 that $\nabla f(x^*) = 0$, hence we only need to prove that $\nabla^2 f(x^*)$ is positive semidefinite. Consider the sequence

$$x_k = x^* + \alpha_k d, \quad \alpha_k > 0,$$

where d is arbitrary. Since $f \in C^2$ and $\nabla f(x^*) = 0$, then by Taylor's expansion, we have for k sufficiently large that

$$0 \leq f(x_k) - f(x^*) = \frac{1}{2} \alpha_k^2 d^T \nabla^2 f(\eta_k) d,$$

where η_k is a convex combination of x_k and x^* . Dividing by $\frac{1}{2} \alpha_k^2$ and taking the limit, we get

$$d^T \nabla^2 f(x^*) d \geq 0, \quad \forall d \in R^n.$$

Hence we complete the proof. $\square$

[proof II] (By contradiction). Suppose that $\nabla^2 f(x^*)$ is not positive semidefinite, then we can choose $d \in R^n$ such that $d^T \nabla^2 f(x^*) d < 0$. Since $f \in C^2$,

there exists $\delta > 0$ and we can choose $\epsilon > 0$ such that $x^* + \epsilon d \in B(x^*, \delta)$ and $d^T \nabla^2 f(x^* + \epsilon d) d < 0$.

By use of $\nabla f(x^*) = 0$, it follows that

$$f(x^* + \epsilon d) = f(x^*) + \frac{1}{2} \epsilon^2 d^T \nabla^2 f(x^* + \theta \epsilon d) d,$$

where $0 \leq \theta \leq 1$. Therefore $f(x^* + \epsilon d) < f(x^*)$. This contradicts the assumption that x^* is a local minimizer. $\square$

Next, we describe the second-order sufficient condition.

Theorem 1.4.6 (*Second-Order Sufficient Condition*)

Let $f : D \subset R^n \rightarrow R$ be twice continuously differentiable on an open set D . If $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is positive definite, then $x^* \in D$ is a strict local minimizer.

Proof. [proof I] Assume that $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is positive definite. By Taylor's expansion, for any vector $d \in R^n$ such that $x^* + d$ lies in a neighborhood of x^* in which $\nabla^2 f(x^* + d)$ is positive definite, we have

$$f(x^* + d) = f(x^*) + \frac{1}{2} d^T \nabla^2 f(x^* + \theta d) d,$$

where $\theta \in (0, 1)$. Then we can choose $\delta > 0$ such that $x^* + d \in B(x^*, \delta)$ and $d^T \nabla^2 f(x^* + \theta d) d > 0$. Therefore,

$$f(x^* + d) > f(x^*)$$

which shows our result. $\square$

[proof II] (By contradiction). Assume that x^* is not a strict local minimizer, then there exists a sequence $\{x_k\} \subset D$ with $x_k \neq x^*$, $\forall k$, such that $f(x_k) \leq f(x^*)$ for k sufficiently large. By use of Taylor's expansion,

$$\begin{aligned} 0 &\geq f(x_k) - f(x^*) \\ &= \nabla f(x^*)^T (x_k - x^*) + \frac{1}{2} (x_k - x^*)^T \nabla^2 f(\eta_k) (x_k - x^*), \end{aligned}$$

where η_k is a convex combination of x_k and x^* . Using $\nabla f(x^*) = 0$, dividing by $\frac{1}{2} \|x_k - x^*\|^2$ and taking the limit, we have

$$0 \geq \bar{e}^T \nabla^2 f(x^*) \bar{e}, \tag{1.4.4}$$

where $\bar{e}$ is the accumulation point of the uniformly bounded sequence $\{(x_k - x^*)/\|x_k - x^*\|\}$ and $\|\bar{e}\| = 1$. Obviously, (1.4.4) contradicts the positive definiteness of $\nabla^2 f(x^*)$. $\square$

Definition 1.4.7 *A point $x^* \in R^n$ is said to be a stationary (or critical) point for the differentiable f if $\nabla f(x^*) = 0$.*

Theorem 1.4.4 tells us that if x^* is a local minimizer, then it is a stationary point. However, the converse is not true. If x^* is a stationary point, it is possible for x^* to be a local minimizer or maximizer, it is also possible for x^* to not be an extreme point. If a stationary point x^* is neither minimizer nor maximizer, it is called a saddle point. Therefore, a stationary point need not be a local minimizer. But if the objective function which is differentiable is convex, its stationary points are the local minimizers and the global minimizers.

Theorem 1.4.8 below says, for a convex function, that its local minimizer is also a global minimizer. Theorem 1.4.9 says, for a differentiable convex function, that its stationary point is also a global minimizer.

Theorem 1.4.8 *Let $S \subset R^n$ be a nonempty convex set and $f : S \subset R^n \rightarrow R$. Let $x^* \in S$ be a local minimizer for $\min_{x \in S} f(x)$.*

1. *If f is convex, then x^* is also a global minimizer.*
2. *If f is strictly convex, then x^* is a unique global minimizer.*

Proof. (1) Let f be convex and x^* be a local minimizer, then there exists a δ -neighborhood $B(x^*, \delta)$ such that

$$f(x) \geq f(x^*), \forall x \in S \cap B(x^*, \delta). \quad (1.4.5)$$

By contradiction, suppose that x^* is not a global minimizer. Then we can find some $\hat{x} \in S$ such that $f(\hat{x}) < f(x^*)$. By convexity of f , we have for $\alpha \in (0, 1)$,

$$\begin{aligned} f(\alpha\hat{x} + (1-\alpha)x^*) &\leq \alpha f(\hat{x}) + (1-\alpha)f(x^*) \\ &< \alpha f(x^*) + (1-\alpha)f(x^*) \\ &= f(x^*). \end{aligned} \quad (1.4.6)$$

But for sufficiently small $\alpha > 0$, $\alpha\hat{x} + (1 - \alpha)x^* \in S \cap B(x^*, \delta)$. Therefore, (1.4.6) contradicts (1.4.5). This contradiction proves the first conclusion.

(2) From part (1) we have that x^* is a global minimizer because strict convexity means convexity. Therefore, it is enough to prove the uniqueness.

By contradiction, suppose that x^* is not the unique global minimizer, so that we can find $x \in S, x \neq x^*$, such that $f(x) = f(x^*)$.

By strict convexity of f ,

$$f\left(\frac{1}{2}x + \frac{1}{2}x^*\right) < \frac{1}{2}f(x) + \frac{1}{2}f(x^*) = f(x^*). \quad (1.4.7)$$

Note from the convexity of S that $\frac{1}{2}x + \frac{1}{2}x^* \in S$. Therefore, (1.4.7) contradicts the fact that x^* is a global minimizer. $\square$

Theorem 1.4.9 *Let $f : R^n \rightarrow R$ be a differentiable convex function. Then x^* is a global minimizer if and only if $\nabla f(x^*) = 0$.*

Proof. Sufficiency. Let f be a differentiable convex function in R^n and $\nabla f(x^*) = 0$, then

$$f(x) \geq f(x^*) + \nabla f(x^*)(x - x^*) = f(x^*), \quad \forall x \in R^n$$

which indicates that x^* is a global minimizer of f .

Necessity. It is obvious because the global minimizer is also a local minimizer, and is also a stationary point. $\square$

The optimality conditions of constrained optimization will be discussed in Chapter 8.

1.5 Structure of Optimization Methods

Usually, the optimization method is an iterative one for finding the minimizer of an optimization problem. The basic idea is that, given an initial point $x_0 \in R^n$, one generates an iterate sequence $\{x_k\}$ by means of some iterative rule, such that when $\{x_k\}$ is a finite sequence, the last point is the optimal solution of the model problem; when $\{x_k\}$ is infinite, it has a limit point which is the optimal solution of the model problem. A typical behavior of an algorithm which is regarded as acceptable is that the iterates x_k move steadily towards the neighborhood of a local minimizer x^* , and then rapidly converge to the point x^* . When a given convergence rule is satisfied, the

iteration will be terminated. In general, the most natural stopping criterion is

$$\|\nabla f(x_k)\| \leq \delta, \quad (1.5.1)$$

where δ is a prescribed tolerance. If (1.5.1) is satisfied, it implies that the gradient vector $\nabla f(x_k)$ tends to zero and the iterate sequence $\{x_k\}$ converges to a stationary point.

Let x_k be the k -th iterate, d_k k -th direction, α_k k -th steplength factor. Then the k -th iteration is

$$x_{k+1} = x_k + \alpha_k d_k. \quad (1.5.2)$$

We can see from (1.5.2) that the different stepsize α_k and different direction d_k form different methods. In Chapter 2 we will discuss several methods to determine α_k . In Chapter 3 we will present various methods to find search directions d_k . Most optimization methods are so-called descent methods in the sense that f satisfies at each iteration

$$f(x_{k+1}) = f(x_k + \alpha_k d_k) < f(x_k),$$

in which d_k is a descent direction defined by Definition 1.4.3.

The basic scheme of optimization methods is as follows.

Algorithm 1.5.1 (*Basic Scheme*)

Step 0. (Initial step) Given initial point $x_0 \in R^n$ and the tolerance $\epsilon > 0$.

Step 1. (Termination criterion) If $\|\nabla f(x_k)\| \leq \epsilon$, stop.

Step 2. (Finding the direction) According to some iterative scheme, find d_k which is a descent direction.

Step 3. (Line search) Determine the stepsize α_k such that the objective function value decreases, i.e.,

$$f(x_k + \alpha_k d_k) < f(x_k).$$

Step 4. (Loop) Set $x_{k+1} = x_k + \alpha_k d_k$, $k := k + 1$, and loop to Step 1. $\square$

Convergence rate, which is a local characterization of an algorithm, can measure the effectiveness of an optimization method. We now give a brief description associated with different types of convergence rate. More details can be found in Ortega and Rheinboldt (1970).

Let the iterate sequence $\{x_k\}$ generated by an algorithm converge to x^* in some norm, i.e.,

$$\lim_{k \rightarrow \infty} \|x_k - x^*\| = 0. \quad (1.5.3)$$

If there are real number $\alpha \geq 1$ and a positive constant β which is independent of the iterative number k , such that

$$\lim_{k \rightarrow \infty} \frac{\|x_{k+1} - x^*\|}{\|x_k - x^*\|^\alpha} = \beta, \quad (1.5.4)$$

we say that $\{x_k\}$ has α -order of Q-convergence rate, where Q-convergence rate means Quotient-convergence rate. In particular,

1. when $\alpha = 1$ and $\beta \in (0, 1)$, the sequence $\{x_k\}$ is said to converge Q-linearly;
2. when $\alpha = 1$ and $\beta = 0$, or $1 < \alpha < 2$ and $\beta > 0$, the sequence $\{x_k\}$ is said to converge Q-superlinearly;
3. when $\alpha = 2$, we say that $\{x_k\}$ has Q-quadratic convergence rate.

The primary motivation for introducing the Q-convergence rate is to compare the speed of convergence of different iterations. It is not difficult to see that the convergence rate depends on α and (more weakly) on β . Suppose that there are two sequences $\{x_k\}$ and $\{x'_k\}$ and that their Q-order and Q-factor are respectively $\{\alpha, \beta\}$ and $\{\alpha', \beta'\}$. If $\alpha > \alpha'$, then the sequence with Q- α order converges faster than the sequence with Q- α' order. For example, a quadratically convergent sequence will eventually converge faster than linearly and superlinearly convergent sequences. When $\alpha = \alpha'$, i.e., their Q-order of convergence rate is the same, if $\beta < \beta'$, then the sequence $\{x_k\}$ is faster than $\{x'_k\}$.

Mainly, we are concerned with Q-linear, Q-superlinear and Q-quadratic convergence. Usually, if the convergence rate of an algorithm is Q-superlinear or Q-quadratic, we say that it has rapid convergence rate. For example, quasi-Newton methods converge Q-superlinearly, and Newton's method converges Q-quadratically.

Another measure of the convergence rate which is weaker than Q-convergence rate is R-convergence rate which means Root-convergence rate.

Let $\{x_k\} \subset R^n$ be any sequence that converges to x^* . Let

$$R_p = \begin{cases} \limsup_{k \rightarrow \infty} \|x_k - x^*\|^{1/k}, & \text{if } p = 1; \\ \limsup_{k \rightarrow \infty} \|x_k - x^*\|^{1/p^k}, & \text{if } p > 1. \end{cases}$$

If $R_1 = 0$, $\{x_k\}$ is said to be R-superlinearly convergent to x^* .

If $0 < R_1 < 1$, $\{x_k\}$ is said to be R-linearly convergent to x^* .

If $R_1 = 1$, $\{x_k\}$ is said to be R-sublinearly convergent to x^* .

Similarly, if $R_2 = 0, 0 < R_2 < 1, R_2 \geq 1$ respectively, then $\{x_k\}$ is said to be R-superquadratically, R-quadratically, and R-subquadratically convergent to x^* respectively.

The above R -rate of convergence can also be stated as follows:

The sequence $\{x_k\}$ is said to be R -linearly convergent if there is a sequence of nonnegative scalars $\{q_k\}$ such that

$$\|x_k - x^*\| \leq q_k \quad \text{for all } k, \text{ and } \{q_k\} \text{ converges } Q\text{-linearly to zero.}$$

Similarly, the sequence $\{x_k\}$ is said to be R -superlinearly convergent if $\{q_k\}$ converges Q -superlinearly to zero; the sequence $\{x_k\}$ is said to be R -quadratically convergent if $\{q_k\}$ converges Q -quadratically to zero.

Similar to Q-rate of convergence, R-rate of convergence also depends on R-order p and R-factor R_p . The higher the R-order is, the faster the corresponding sequence converges. When the R-order is the same, the smaller the R-factor is, the faster the corresponding sequence converges.

Throughout this book we mainly discuss Q-convergence rate. Hence, if there is not specific indication, the convergence rate refers to Q-convergence rate.

As indicated above, usually an algorithm with superlinear or quadratic rate is said to be desirable. However, it must be appreciated that the theoretical results of the convergence and convergence rate are not a guarantee of good performance. Not only do these results themselves fall short of guarantee of good behavior, but also they neglect computer round-off errors which may be crucial. In addition, these results often impose certain restrictions on $f(x)$ which may not be easy to verify, and in some cases (for example, in the convex case), these conditions may not be satisfied in practice. Thus,

the development of an optimization method also relies on numerical experimentation. The ideal is a good selection of experimental testing backed up by the proofs of convergence and convergence rate.

We have known from the above discussion that the convergence rate measures the local behavior of an algorithm and is used in local analysis. The theorem below gives a characterization of superlinear convergence which is useful for constructing termination criteria.

Theorem 1.5.2 *If the sequence $\{x_k\}$ converges Q-superlinearly to x^* , then*

$$\lim_{k \rightarrow \infty} \frac{\|x_{k+1} - x_k\|}{\|x_k - x^*\|} = 1. \quad (1.5.5)$$

However, in general, the converse is not true.

Proof. For a given integer $k \geq 0$,

$$\begin{aligned} \frac{\|x_{k+1} - x^*\|}{\|x_k - x^*\|} &= \frac{\|(x_{k+1} - x_k) + (x_k - x^*)\|}{\|x_k - x^*\|} \\ &\geq \left| \frac{\|x_{k+1} - x_k\|}{\|x_k - x^*\|} - \frac{\|x_k - x^*\|}{\|x_k - x^*\|} \right|. \end{aligned}$$

It follows from the definition of Q-superlinear convergence that

$$\lim_{k \rightarrow \infty} \frac{\|x_{k+1} - x_k\|}{\|x_k - x^*\|} = 1.$$

To show that the converse is not true, we give a counter-example. In normed space $\{R, |\cdot|\}$, define a sequence $\{x_k\}$ as follows:

$$\begin{aligned} x_{2i-1} &= \frac{1}{i!} \quad (i = 1, 2, \dots), \\ x_{2i} &= 2x_{2i-1} \quad (i = 1, 2, \dots). \end{aligned}$$

Obviously, $x^* = 0$. We have

$$\frac{|x_{k+1} - x_k|}{|x_k - x^*|} = \begin{cases} 1, & k = 2i - 1, i \geq 1, \\ 1 - \frac{1}{2(i+1)}, & k = 2i, i \geq 1. \end{cases}$$

So, $\{x_k\}$ satisfies (1.5.5) but does not converge Q-superlinearly to x^* . $\square$

This theorem shows that if an algorithm is convergent Q-superlinearly, instead of $\|x_k - x^*\|$, we can use $\|x_{k+1} - x_k\|$ to give a termination criterion, and the estimation will be improved as k increases.

Finally, we discuss some termination criteria which are used frequently in practice. In order to guarantee convergence of an algorithm, we require

$$|f(x_k) - f(x^*)| \leq \epsilon \text{ or } \|x_k - x^*\| \leq \epsilon,$$

where the parameter ϵ is user-supplied. Unfortunately, these are not practicable since they need the information of the solution x^* .

Instead, we often use the following termination criteria:

$$\|\nabla f(x_k)\| \leq \epsilon_3, \tag{1.5.6}$$

$$\|x_{k+1} - x_k\| \leq \epsilon_1, \tag{1.5.7}$$

$$f(x_k) - f(x_{k+1}) \leq \epsilon_1. \tag{1.5.8}$$

Normally, when an algorithm can be expected to converge rapidly, it is suggested to use (1.5.7) or (1.5.8). When an algorithm has first-order derivative information and can be expected to converge less rapidly, a test based on (1.5.6) may be appropriate.

Himmeblau [174] suggested that it is suitable to use (1.5.7) together with (1.5.8) as follows:

When $\|x_k\| > \epsilon_2$ and $|f(x_k)| > \epsilon_2$, use

$$\frac{\|x_{k+1} - x_k\|}{\|x_k\|} \leq \epsilon_1, \quad \frac{|f(x_k) - f(x_{k+1})|}{|f(x_k)|} \leq \epsilon_1; \tag{1.5.9}$$

otherwise, use

$$\|x_{k+1} - x_k\| \leq \epsilon_1, \quad |f(x_k) - f(x_{k+1})| \leq \epsilon_1. \tag{1.5.10}$$

He also suggested using (1.5.9)-(1.5.10) together with (1.5.6).

In general, take $\epsilon_1 = \epsilon_2 = 10^{-5}$, $\epsilon_3 = 10^{-4}$.

Exercises

1. Let A be an $n \times n$ nonsingular matrix. Prove that $\|Ax\| \geq \|x\|/\|A^{-1}\|$.
2. Prove the equivalence (1.2.22)-(1.2.26) of vector norms.

3. Prove Cauchy-Schwarz inequality (1.2.34). Further, prove inequality (1.2.35).

4. Prove (1.2.45).

5. Let $A = UDV^*$ be the singular value decomposition. Prove that $A^+ = VD^+U^*$, where D^+ is defined in (1.2.54).

6. Prove Sherman-Morrison formula (1.2.67) and Sherman-Morrison-Woodburg formula (1.2.68).

7. Show Theorem 1.2.6 (Von-Neumann Lemma).

8. Prove (1.2.69) and (1.2.70).

9. Prove that a function that is Fréchet differentiable must be Gateaux differentiable, but the converse is not true.

10. Prove Theorem 1.2.23.

11. Show that the intersection of finitely many convex sets is a convex set.

12. Show by induction that the set $S \subset R^n$ is convex if and only if for any $x_1, x_2, \dots, x_m \in S$,

$$\sum_{i=1}^m \alpha_i x_i \in S,$$

where $\sum_{i=1}^m \alpha_i = 1, \alpha_i \geq 0, i = 1, \dots, m$. That means a convex combination of arbitrarily finitely many points of a convex set still belongs to the convex set.

13. Let $A \in R^{m \times n}, b \in R^m$. Show, by definition, that

$$S = \{x \in R^n \mid Ax = b, x \geq 0\}$$

is a convex set.

14. Let

$$D_1 = \{x \mid x_1 + x_2 \leq 1, \quad x_1 \geq 0\}, \quad D_2 = \{x \mid x_1 - x_2 \geq 0, \quad x_1 \leq 0\}.$$

Set $D = D_1 \cup D_2$. Show that D is not necessarily convex though both D_1 and D_2 are convex. This means that the union of convex sets is not necessarily a convex set.

15. Write the convex hull of the set $S = \{(0,0)^T, (1,0)^T, (0,1)^T\}$.

16. Let $S \subseteq R^n$. Prove that the following two statements are equivalent.

- (1) The convex hull is the set of all convex combinations of arbitrarily finitely many elements of S as defined in (1.3.3).
- (2) The convex hull $\text{conv}(S)$ is the intersection of all convex sets containing S .

17. Let $f_i(x), i = 1, 2, \dots, m$, be convex functions defined on convex set $D \subset R^n$. Show that the function

$$g(x) = \sum_{i=1}^m \alpha_i f_i(x)$$

is also a convex function on D , where $\sum_{i=1}^m \alpha_i = 1, \alpha_i \geq 0, i = 1, 2, \dots, m$. This means that the convex combination of convex functions is a convex function.

18. Discriminate convexity of the following functions:

- (1) $f(x_1, x_2) = x_1 e^{-(x_1+x_2)}$;
- (2) $f(x_1, x_2, x_3) = x_1^2 + 3x_2^2 + 9x_3^2 - 2x_1x_2 + 6x_2x_3 + 2x_3x_1$.

19. Prove Theorem 1.3.11.

20. State the first-order and second-order optimality conditions for unconstrained optimization and outline their proofs.